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REVIEW ARTICLE

Recent Progress in PEG-Based Composite Phase Change Materials

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ABSTRACT

This review discusses advances in polyethylene glycol-based composite phase change materials (PCMs) for thermal energy storage (TES) and thermal regulation. PCMs utilize latent heat storage, absorbing and releasing energy during phase transitions within specific temperature ranges. Polyethylene glycol (PEG) is a promising organic PCM due to its easily tunable phase change temperatures, high melting/freezing enthalpies, and nontoxicity, among other advantages. However, PEG suffers from low thermal conductivity and requires encapsulation to contain the flow of liquified PEG. To address these issues, PEG has been composited with thermally conductive fillers and porous materials. Moreover, PEG has been modified to have enhanced photothermal conversion efficiency, decreased supercooling, and flame resistance. This review discusses exemplary developments in PEG-based composite PCMs, focusing on blending with different polymers, doping with various carbon materials (porous carbons, graphene, and carbon nanotubes), embedding into silica-based skeletons, and synergizing with other promising hosts and additives like layered doubled hydroxides, MXenes, and metal-organic frameworks. This work highlights key studies focused on implementing PEG-based PCMs in building, pavement, electronic, textile, solar, and waste heat recovery applications. The consequences of different synthesis parameters and their effects on the composite PCM's thermal transition properties are emphasized among the other results.

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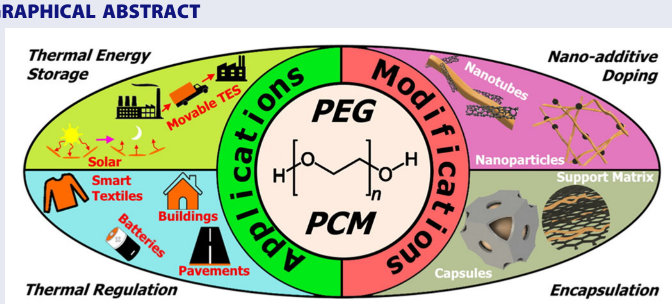
Phase change materials; polyethylene glycol; carbon nanomaterials; silica; thermal energy storage; thermal regulation

GRAPHICAL ABSTRACT



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1. Introduction

Growing energy demands and increasing environmental strain have prompted the shift toward cleaner energy sources and improved energy management.[1,2] However, most renewable energy sources, like solar and wind, require excess energy to be stored in preparation for sudden energy demand spikes and in case the renewable resource becomes unavailable, like during cloud coverage or calm weather. Thermal energy storage (TES) systems act as a buffer against energy supply/demand fluctuations, storing surplus heat and releasing heat when in demand.[3,4] While some TES systems use sensible heat[5–7] or sorption (thermochemical) heat[8–10] storage mechanisms, TES systems that use latent heat storage (phase change energy storage) are promising because of their high energy densities and minimal temperature changes during energy storage and release.[11–13] The critical components in latent heat storage systems are the phase change materials—inorganic,[14–17] organic,[18–21] eutectic,[22] or composite[23,24] materials that absorb and release large amounts of heat during phase changes.[25,26] PCMs store energy from various renewable or waste heat sources, including solar irradiation,[27] body heat,[28] industrial waste heat,[29] and electronic systems.[30] PCMs are also commonly used for thermal regulation, wherein the PCM reduces the effect of temperature fluctuations by absorbing excess heat or releasing heat in colder conditions.[31]

Ideal PCMs are materials with large latent heat, high thermal conductivity, minimum density change during phase transitions, chemical stability, cyclic stability, thermal stability, and low cost.[2] Depending on the application, PCMs may require high flexibility, mechanical strength, solvent resistance, flame resistance, nontoxicity, and chemical inertness, among other traits. Common inorganic PCMs are salt hydrates, such as $CaCl_2{\cdot}6H_2O$,[32] $Na_2SO_4{\cdot}10H_2O$,[33] and $Na_2CO_3{\cdot}10H_2O$,[34] which have high latent heat, good thermal conductivity, and low cost.[17] However, inorganic PCMs are corrosive and exhibit significant supercooling, phase segregation, and noncongruent melting.[14–16] Conversely, organic PCMs like paraffin waxes (PWs), glycols, fatty acids, alcohols, and lipids exhibit minimal supercooling, small density changes during phase transition, no phase separation, and are non-corrosive.[18–20] However, organic PCMs have low thermal conductivities and experience significant leakage, leading to a permanent loss of PCM material after each phase change cycle.[18,19] Eutectic PCMs are organic-organic, inorganic-inorganic, or organic-inorganic PCM mixtures with synergistic effects like lower melting temperatures and higher latent heat than their constituents.[22–24]

PEG is a promising organic PCM with high phase change enthalpies and easily tunable phase transition temperature ranges.[35,36] Other advantages of PEG-based PCMs include facile synthesis, nontoxicity, biocompatibility, low cost, flexibility, compatibility with various additives, and potential for fabrication without catalyst or harmful solvent.[37] Because of such advantages, PEG has been used in many other applications, including polymer modifications,[38–42] photovoltaics,[43–46] solar concentrators,[47] separation membranes,[48–50] and antifouling membranes,[51] and drug delivery,[52] amongst various other applications. As illustrated in Figure 1, PEG PCMs can be used for various thermal regulation applications, such as in buildings, pavements, batteries, textiles, and TES systems like solar and movable TES. Still, there are two critical issues with PEG PCMs to address. First, the thermal conductivity of PEG must be increased to

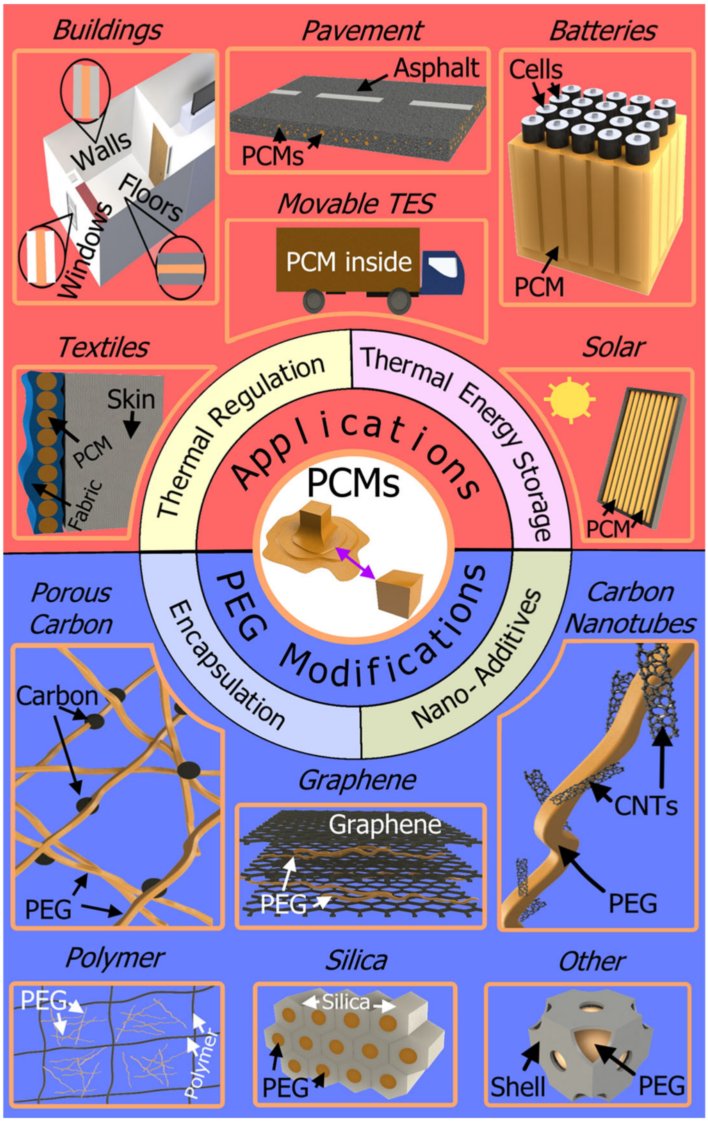

Figure 1. Schematic summarizing the (top) different applications of PEG-based composite PCMs and (bottom) the main PEG modifiers emphasized in this review.

enable better heat transfer efficiency. Second, PEG-based PCMs should be modified to have solid-solid phase transitions instead of their natural solid-liquid phase transition. These so-called "form-stable" PEG-based PCMs eliminate the need for external containers, reducing the cost, weight, and complexity of applying PEG PCMs to thermal regulation and TES applications.

The latest PCM reviews only briefly discuss PEG as a type of organic PCM[53,54] and only highlight a few applications of PEG-based PCMs.[55,56] However, PEG deserves to be emphasized as a competitive PCM in its own comprehensive review, especially with the latest improvements involving thermally conductive additives and highly porous nanomaterials. Thus, this review focuses on PEG-based composite PCMs, emphasizing modifiers that increase thermal conductivity, endow form stability, and add functionality for specific PCM applications. As shown at the bottom of Figure 1, this review discusses the performance of PEG-based composite PCMs modified with various polymers, carbons (1D carbon nanotubes, 2D graphene, and 3D porous carbons), silica, and other hosts and dopants with high thermal conductivity. The synthesis parameters to consider and optimize when synthesizing PEG-based composite PCMs are analyzed with differential scanning calorimetry (DSC) and selected results from exemplary studies. Potential applications of PEG-based composite PCMs are summarized, highlighting the optimal PEG thermal properties. Lastly, the future direction of PCMs is discussed. This paper hopes to show the advantages and limitations of PEG-based composite PCMs for various thermal regulation and TES applications and guide further research in enhancing PEG-based PCMs for widespread adoption.

2. Pristine PEG PCMs

The thermal properties of PEG are highly dependent on its molecular weight (MW). For example, PEG's melting temperature (T_m) increases from 6 °C for 400 g/mol, to 36 °C for 2000 g/mol, and to 58 °C for 10,000 g/mol.[36,57] The energy density of PCMs is typically measured by specific enthalpies (latent heat) of melting (ΔH_m) and crystallization (ΔH_c), which increases with MW. For example, PEG-1000 has a ΔH_m of 165.3 J/g, while PEG-35000 has a 21% higher enthalpy of 199.7 J/g.[58] PEG's thermal conductivity also generally increases with MW, increasing from 0.16 W/mK for PEG-1000, to 0.19 W/mK for PEG-6000, and to 0.24 W/mK for PEG-10000.[35,36]

One of the largest drawbacks of PEG is its low thermal conductivity (below 0.3 W/mK). For perspective, the thermal conductivities of inorganic PCMs like $CaCl_2{\cdot}6H_2O$ (0.56 W/mK)[59] and $Na_2SO_4{\cdot}10H_2O$ (0.89 W/mK)[60] are around 2 and 3 times higher, respectively. Graphene (4000 W/mK)[61] has over 10,000-times higher thermal conductivity. High thermal conductivity is important because it ensures the congruent freezing/melting of PEG and the efficient thermal energy transfer from the heat source. Moreover, applications like pavements and textiles that experience heat concentrated on one side require high thermal conductivity to distribute heat quickly throughout the bulk material. High thermal conductivity also improves thermal stability by dispersing heat before the thermal degradation threshold is reached in a particular area. To improve the thermal conductivity of PEG, various thermally conductive materials such as graphene,[62] carbon nanotubes (CNTs),[63] and silica[64] have been composited with PEG as a dopant or host material. However, PEG-based composite PCMs have limited PEG content, resulting in lower enthalpies of melting and freezing than pristine PEG.[65] Thus, PEG composite PCMs strive to maximize PEG loading with satisfactory thermal conductivity.

The disadvantage of a PEG PCM's solid-liquid phase transition is leakage, which is when liquid PEG flows out of the PCM system and is irrecoverably lost.[66] While a solid-liquid transition is preferable to a liquid-gas transition for most applications because solid-liquid transitions have minimal density changes,[67] solid-liquid transition PCMs require external components to contain the melted PCM. The external encapsulation adds to the cost, weight, and volume of the TES or thermal regulation system and makes applying the PCMs to various applications more complex. Thus, solid-solid transition PEG (form-stable PEG), which does not exhibit leakage, is ideal for most applications. The primary method of endowing form stability to PEG is *via* the immobilization of PEG within the pores of various materials, including other polymers,[68] mesoporous silica,[69] and porous carbon.[70] Encapsulation, entrapment, and impregnation are used interchangeably in this review to refer to the process of immobilizing PEG within the host's pores to prevent leakage. As shown in Figure 2, complete microcapsule-type encapsulation can prevent PEG leakage by completely encasing PEG to prevent the flow of PEG macromolecules. Impregnating porous hosts with PEG also reduces leakage through strong intermolecular interactions between the host and PEG. However, using a host material necessarily lowers the thermal storage capacity due to decreased PEG content and disturbed crystallization. Thus, the primary goal of PEG encapsulation studies is to maximize PEG loading while having no PEG leakage even after many phase transition cycles.

3. Composite PEG PCMs

The most common modifications to PEG-based PCMs revolve around improving their low thermal conductivity and endowing form stability. These materials include various polymers, carbon materials (1D CNTs, 2D graphene, and 3D hierarchical polar carbon (e.g. bio-macroporous derived carbon, expanded graphite, and carbon foam)), silica skeletons, and other thermally conductive or porous additives. Blending PEG with other polymers, especially cross-linked polymers, endows form stability with highly tunable thermal properties. Carbon-based materials are popular modifiers for PEG-based PCMs because of their excellent thermal conductivity, photothermal conversion efficiency, and ecofriendliness. Carbon modifiers used for PEG modification can be broken down by dimensionality: 3D hierarchical porous carbons, 2D graphene, and 1D CNTs. Similarly, silica-based skeletons can host PEG to increase PEG's thermal degradation threshold and reduce leakage. Other promising nanomaterials like MXenes, layered-double

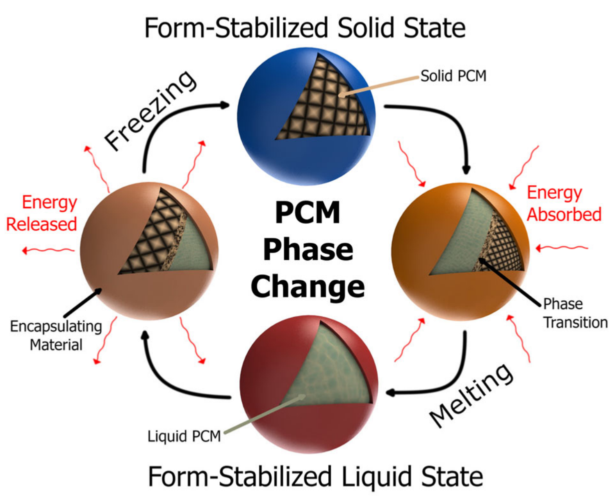

Figure 2. Schematic of a form-stable PEG-based PCM undergoing melting and freezing.

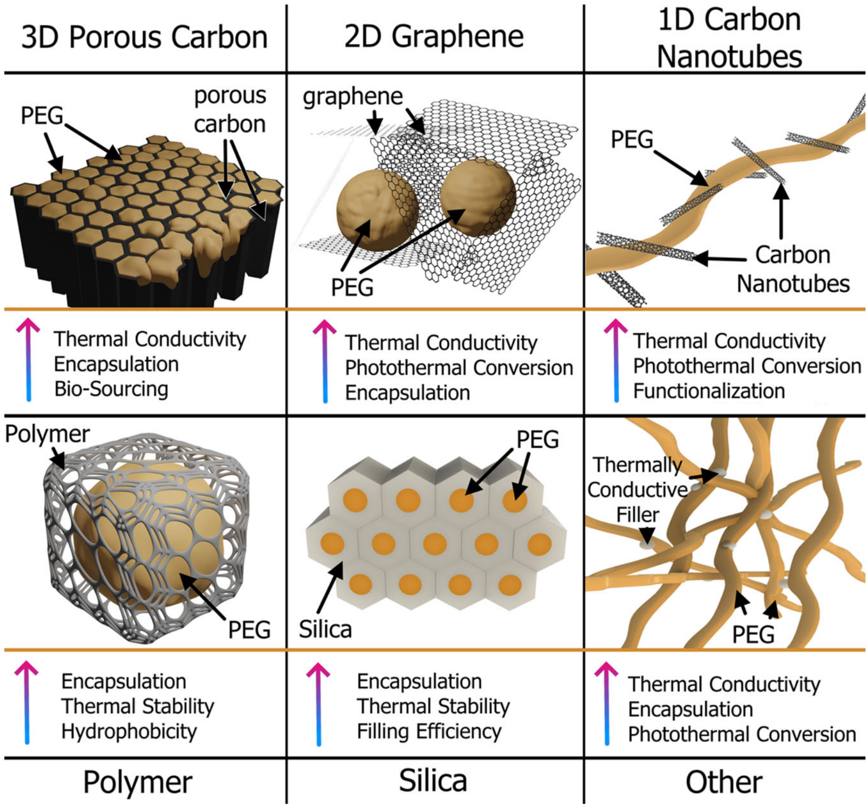

Figure 3. The advantages of modifying PEG with the various encapsulation or nano-additive materials discussed in this review.

hydroxides, metal foams, and metal nanoparticles endow form stability and enhance thermal conductivity. Figure 3 summarizes the benefits of modifying PEG with the polymer, carbon, silica, and other modifiers discussed in this review.

3.1. PEG/polymer composite

The primary objective of blending PEG with different polymers is to endow form stability over many melting/freezing cycles while maximizing the latent heat of the composite PCM.[71] A key parameter to minimize during PEG/polymer PCM fabrication is the pore size of the encapsulating material. Smaller pore sizes result in reduced PEG leakage because smaller pores trap the long-chain PEG from flowing out. For example, Samani et al. found that higher Novolac resin content yielded composite PCMs with smaller pore sizes.[72] The smaller pore sizes were able to fully entrap the PEG chains to reduce leakage. However, the resin could not form a contiguous pore structure due to the interference of clay particles in the composite PEG/Novolac/clay composite PCM, resulting in gaps that caused PEG leakage. Thus, in addition to small pore sizes, the overall pore structure must be designed to have interconnected channels to fully encapsulate the PEG.

PEG leakage can be controlled by adjusting the degree of cross-linking in the encapsulating polymer. In recent work, Wang et al. compared the thermal performance of PEG encapsulated with linear versus cross-linked polyurea.[73] PEG with cross-linked polyurea (PEG/CPU) had a higher PEG encapsulation efficiency of 75 wt%, whereas PEG/linear polyurea (PEG/LPU) had a lower encapsulation rate of 65 wt%. Cross-linked polyurea had smaller pore sizes and stronger intermolecular interactions with PEG, resulting in higher encapsulation and reduced leakage during melting/freezing cycles. Comparing the DSC thermographs for PEG/LPU (Figure 4a) and PEG/CPU (Figure 4b), PEG/CPU has a wider heat flow curve than PEG/LPU and a lower peak heat flow, resulting in PEG/CPU having a lower specific latent heat than PEG/LPU at the same encapsulation efficiency of 65 wt% PEG. However, the overall latent heat was higher for PEG/CPU than PEG/LPU because PEG/CPU had a higher maximum encapsulation without leakage of 75 wt% PEG (Figure 4c).

A high degree of cross-linking can disrupt crystallization, reducing latent heat.[74] For example, Wu et al. found that ΔH_m and ΔH_c were 33% lower for the cross-linked composite than the control PEG, partially due to crystallization interference.[74] As shown in Figure 4(d), 9,10-dihydro-9-oxa-10-phosphaphenanthrene-10-oxide (DMEP) was cross-linked with Jeffamine T-403 in the presence of PEG, resulting in the PEG hosted within a cross-linked polymer network. As expected, the cross-linked PEG composites had excellent form stability and minimal leakage up to 75 wt% PEG. However, the degree of hysteresis (difference between melting and freezing temperatures) increased due to the cross-linked DMEP preventing PEG nucleation.

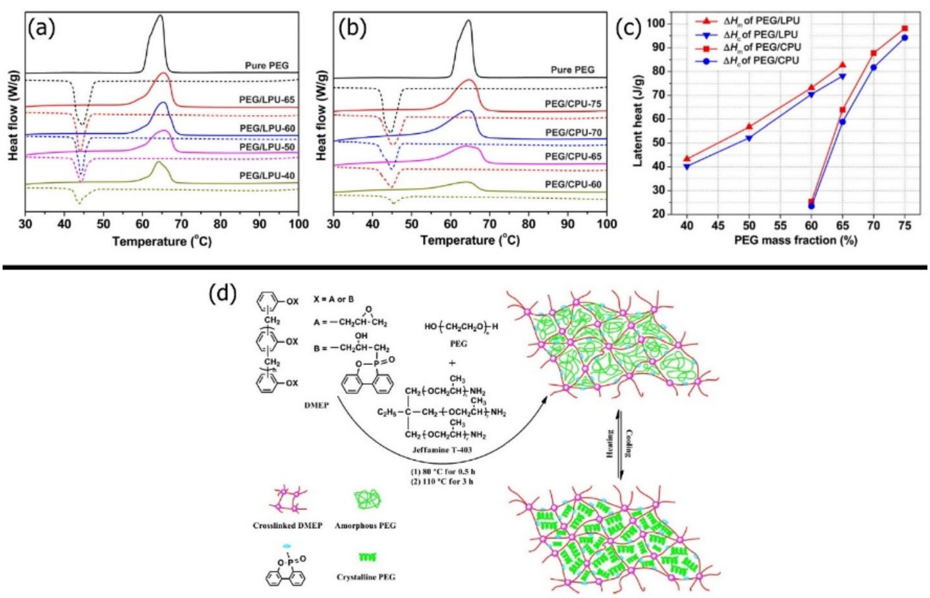

Figure 4. DSC thermographs of pristine PEG, (a) PEG/LPU and (b) PEG/CPU with varying wt% PEG. (c) Latent heat of melting and freezing for PEG/LPU vs. PEG/CPU.[73] (Reproduced with permission from [73], Copyright Wiley-VCH, 2022). (d) Schematic for the preparation of PEG/DMEP composites and phase change.[74] (Reproduced with permission from [74], Copyright Elsevier, 2018).

One method to tune the cross-linking density is by cross-linking the host polymer with additives that possess natural cavities, such as β-cyclodextrin (β-CD). Zhou et al. found that cross-linking ethylene-propylene-diene (EPDM) with β-CD yielded a 156% higher ΔH_m than the PEG/EPDM composite without β-CD.[75] This was because the more porous PEG/EPDM/β-CD had less restricted PEG crystallization. The β-CD also improved the chemical affinity between the EPDM support matrix and polar PEG, which further increased encapsulation efficiency by 250% compared to PEG/EPDM.

Instead of combining PEG with cross-linked polymers, PEG can be cross-linked directly with the support polymer. Peng et al. fabricated a form-stable PEG-based composite PCM by cross-linking PEG with β-CD and 4,40-diphe-nylmethane diisocyanate (MDI) *via* thermal curing.[76] The composites were prepared with varying molar ratios of β-CD, resulting in form-stable PEG composites with different cross-linking densities. DSC showed that the ΔH_m and ΔH_c of the PCMs increased proportionally with increasing β-CD content because higher β-CD content decreased the cross-linking density, resulting in less PEG chain confinement. Similarly, Kong et al. used a solvent-free method to create a solid-state PCM comprised of PEG hosted within a PEG/polyurethane (PU) cross-linked polymer blend.[68] The cross-linked structure entrapped excess PEG, and the interfacial bonding between PEG/PU host reduced PEG mobility in the molten phase. Thus, the composite PCM had no leakage for over 100 thermal cycles up to 100 °C. However, the limited PEG encapsulation of 50 wt% resulted in a 32% lower ΔH_m and ΔH_c than pristine PEG. The cross-linked host affected PEG crystallization more in composites with lower PEG content. PEG/PU with 50 wt% PEG exhibited a differential ratio of 1.56% than the theoretically calculated value for ΔH_m because the cross-linked PU disrupted PEG crystallization. However, the PEG/PU composite with 20 wt% PEG had an 11-times-higher differential ratio of 17.0%, suggesting more significant interference effects with lower PEG content.

Although PEG/polymer blends generally decrease latent heat, blending PEG with other polymers may add other functionality, like thermal stability, photothermal conversion efficiency, flame resistance, and aqueous solvent resistance. For example, Huang et al. synthesized a cross-linked PEG/triallyl isocyanurate (PEG/TAIC) composite that had increased thermal stability above 300 °C, sufficient for high-temperature pavement applications.[77] The TAIC actually decreased thermal conductivity slightly by 2% (10 wt% TAIC) up to 5% (25 wt% TAIC). The latent heats of the PEG/TAIC composite were lower by 15% than pure PEG. However, PEG/TAIC had a 15 °C higher thermal degradation onset temperature of 392 °C. In another study, Sundararajan et al. traded a reduction in ΔH_m for significantly higher thermal stability and hydrophobicity *via* encapsulation with cross-hydroxy-terminated poly(dimethyl siloxane) (HTPDMS).[78] While HTPDMS encapsulation decreased the latent heat by 83% due to the low 63 wt% PEG encapsulation with HTPDMS, PEG/HTPDMS had a 34 °C higher thermal degradation onset temperature of 192 °C and a 179 °C higher final degradation temperature of 496 °C. The HTPDMS increased thermal stability by leaving behind a non-degradable and nonvolatile char layer at the surface of the PCM, insulating the underlying material. Polydopamine (PDA) is a photothermal polymer that can enhance the light-to-heat conversion properties of PEG.[79] Du et al. found that the PEG with a PDA-modified melamine foam support material was 5.9 °C hotter than PCM with non-modified melamine foam.[80] Polymer-based host matrices and encapsulation material may also deter PEG combustion. Du et al. studied PEG-based composites utilizing PU as a support matrix and PDA decorated black phosphorous (BP) nanosheets as fillers to enhance both the solar-thermal conversion ability and the flame retardancy of the composite.[81] The solar-thermal conversion and storage efficiency of the composites increased from 63.2 to 88.5% with increasing PDA/BP content. Microcombusion calorimetry and loss on ignition (LOI) testing found improve flame resistance, with a decreased peak heat release rate (pHRR) lower total heat release, increased char yield, and higher LOI. The increased flame retardant capability of the PDA/BP containing composites was mostly attributed to the combustion of surface PDA/BP that formed a char layer that blocked combustible gas diffusion to the inner layers of the composite. In more recent work, Yang et al. encapsulated PEG with poly(acrylamide-co-acrylic acid) copolymer (PAAA) to enable high PEG encapsulation in an aqueous HTF.[82] The abundant carboxyl groups in the PAAA had strong interactions with polar PEG, enabling a high encapsulation of 89 wt% without leakage despite being dispersed in distilled water. High PEG stability in aqueous mediums makes it more suitable for various TES and thermal regulation applications in HTFs.

A unique advantage of PEG/polymer composite PCMs is their ability to be affordably and easily spun into fibers for various applications like smart textiles (Sec. 4.4). One of the main factors to consider when fabricating PEG-based fibers is the natural decrease in tensile strength with high PEG content. Chen et al. found that PEG-based form-stable composite fiber PCMs using cellulose acetate (CA) as an encapsulation material exhibited a 48% lower tensile strength than pure CA fibers.[83] Stress-strain curves for the composites showed that increasing PEG content from 37 wt% PEG to 43 wt% weakened the tensile properties of the fibers from 6 MPa to 5 MPa because the PEG disrupted the continuous phase structure of CA. Hence, high PEG-loaded fabrics may be better used in low-stress applications, such as smart clothing.

Instead of standard electrospinning, other polymer spinning techniques, like centrifugal spinning[84] and melt-blowing,[85] have been used in nanofiber fabrication. Zhang et al. used a novel and facile centrifugal spinning technology to produce composite fibers costing of a PEG PCM and polyvinylpyrrolidone (PVP) support matrix.[86] Unlike electrospinning, centrifugal spinning may be more easily scaled up, as it does not require a high-voltage electric field and is less sensitive to the environment.[86] SEM analysis validated the formation of a fibrous morphology, with lower PEG mass ratios yielding smoother, more cylindrical fibers at the cost of decreased latent heat. Recently, Zhao et al. fabricated a PEG/PU composite PCM into a nonwoven fabric using a melt-blowing process.[87] The resulting PCM fabric had a good tensile strength of 19.75 MPa, high elongation of 620%, and excellent air permeability of 413 g/m^2, making the fabric suitable for textile applications. Like with all spinning techniques, the mass ratio of PEG to the host polymer composite must be optimized for the desirable nanofiber morphology.

One of the advantages of spinning-synthesized PEG/polymer PCMs is the ability to integrate dopants easily for improved PCM properties. For example, Song et al. decorated PEG/polyvinyl alcohol (PVA) nanofibers with Ag nanoparticles (PEG/PVA/Ag) as eco-friendly, form-stable composites.[88] The addition of 15 wt% Ag NPs decreased the

$\Delta H_{m,}$ and ΔH_c by 7% and 10%, respectively, owing to decreased PEG content. However, the thermally conductive Ag NPs made the thermal energy release times about 26% faster than pure PEG. PEG/PVA with 15 wt% Ag NPs also showed excellent cyclability, with a small 6.6% decrease in the heat of melting after 2000 melting/freezing cycles. Other dopants may be added to PCM fibers to increase the functionality of fibrous PEG/polymer PCMs. For example, Shi et al. embedded luminescent europium particles into electrospun PCM composites for solar and sensor applications.[89] Such PEG/polymer/dopant composites show promise as a luminescent TES material used for smart wound dressings, controlled drug delivery systems, and nano-sensing applications.

Overall, PEG modified with various polymer materials yield form-stable with good thermal cyclability and minimal leakage. Depending on the polymer host, the composite PCM may also have higher thermal stability, photothermal conversion efficiency, hydrophobicity, and flame resistance. While cross-linking can decrease pore sizes and better encapsulate PEG, high cross-linking densities can severely limit the latent heat by hindering PEG phase change. Hence, cross-linking densities must be optimized by adding porous dopants or tuning polymer content. Spinning technology is especially useful for applying polymer encapsulated PEG PCMs in smart textiles or other low-stress applications, owing to their ease of fabrication and modification with dopants that increase thermal conductivity and provide other functionality. The performance of various PEG/polymer PCMs is summarized in Table 1.

3.2. PEG/carbon composites

Carbon materials may be more suited as host materials that endow form stability or as dopants that enhance thermal conductivity depending on their dimensionality. Hence, this section separates the discussion of carbon composites into 1D carbon nanotubes, 2D graphene, and 3D porous carbon. Generally, carbon modifiers are used primarily to increase thermal conductivity. Other benefits may include improving photothermal conversion efficiency and thermal stability.

3.2.1. 1D carbon nanotubes

CNTs are 1D carbon nanoparticles that are doped into PEG to increase thermal conductivity. Because CNTs have incredibly high thermal conductivities (2800–6000 W/mK),[91] even small amounts of CNT can significantly increase the thermal conductivity of PEG.[92] For example, Qian et al. found that a small 4 wt% loading of single-walled carbon nanotube (SWCNT) increased the thermal conductivity of solid phase PEG by 375%.[63] Because of the high PEG content, PEG/SWCNT had latent heat nearly identical to pristine PEG. Further increasing SWCNT content to 10 wt% increased the thermal conductivity to 3.43 W/mK, one of the highest thermal conductivities reported for PEG-based PCMs. 1D CNTs are generally better dopants for PEG PCMs than 0D carbon nanoparticles (CNPs). For example, Qian et al. found that 1D CNTs outperformed 0D CNPs, with a 57% higher thermal conductivity.[93] PEG/diatomite with 1D SWCNTs increased the thermal conductivity of PEG/diatomite because the 1D structure

Table 1. Summary of the phase change properties of PEG/polymer composite PCMs.

Composite	Composition	Synthesis	T_m (°C)	T_c (°C)	ΔH_m (J/g)	ΔH_c (J/g)	Thermal conductivity (W/m·K)	Ref.
PEG/DMEP	70/30	Thermal curing	67.7	27.9	112.0	108.0	–	[74]
PEG/EPDM/St/β-CD-MA	7.5/6/4/2	Physical adsorption	61.3	33.1	68.8	61.4	–	[75]
PEG/MDI/β-CD	1/2/(4/7)	Thermal curing	60.2	47.8	115.2	111.6	–	[76]
PEG/TIAC	9/1	In-situ polymerization	57.1	31.2	136.9	135.7	0.149	[77]
PEG/TEOS/HTPDMS	3/1/1	Condensation polymerization	60.9	22.9	116.1	97.2	–	[78]
PEG/MPMF	98 wt% PEG	Vacuum impregnation	62.5	36.0	186.2	175.3	–	[80]
PEG/PU/PDA/BP	99 wt% PEG/PU	Polymerization under N_2 atmosphere	52.3	24.2	127.3	124.4	0.459	[81]
PEG-adipic acid/PBT	91 wt% PEG	Esterification and polycondensation	45.6	33.4	136.8	110.1	–	[90]
PEG/CA	43 wt% PEG	Electrospinning	58.3	42.8	60.6	–	–	[83]
PEG/PVA/Ag	15 wt% Ag	Electrospinning	45.9	31.4	66.5	65.7	–	[88]
PEG/PVP	7/3	Centrifugal Spinning	65.0	–	141.9	–	–	[89]

Acronyms: DMEP (9,10-dihydro-9-oxa-10-phosphaphenanthrene-10-oxide); EPDM (ethylene-propylene-diene); St (styrene); β-CD-MA (β-cyclodextrin with a vinyl group); MDI (4,4-diphenylmethane diisocyanate); β-CD (β-cyclodextrin); TIAC (triallyl isocyanurate); TEOS (tetraethyl orthosilicate); HTPDMS (cross-hydroxy-terminated poly(dimethyl siloxane)); MPMF (PDA modified melamine foam with Mxene nanosheets); PU (polyurethane); PDA (polydopamine); BP (black phosphorous); PBT (poly(butylene terephthalate); CA (cellulose acetate); PVA (polyvinyl alcohol); PVP (polyvinylpyrrolidone).

of the SWCNTs allowed for better packing with PEG in the porous diatomite and more continuous heat transfer.

Similar to SWCNTs, multi-walled CNTs (MWCNTs) may also boost the thermal conductivity of PEG, as demonstrated by Sun et al.[94] Thermal conductivity of PEG-$CaCl_2$/MWCNT composites increased linearly with increasing MWCNT content from 5 to 20%, with a maximum thermal conductivity of 0.91 W/mK (2.5-times-greater than pure PEG-$CaCl_2$). The electrical resistivity of the composites decreased with MWCNT content (5–20%) from 9500 to 91 Ω. Electrothermal conversion and storage efficiency were also measured, where PEG-$CaCl_2$/MWCNT with 20 wt% MWCNTs had an electrothermal efficiency of 70.2% under a voltage of 1.5–2.0V. The ability to convert electrical waste heat efficiently is a facet of PCMs that should be studied in more detail. In a more recent study, Cui et al. decreased the heating and cooling times of cross-linked PEG thin film PCM by adding 2 wt% MWCNTs.[95] Heating from 24 to 90 °C took 394 s for pristine PEG but only 44 s for PEG/MWCNTs, owing to the significantly higher thermal conductivity.

One of the most promising attributes of CNTs is their ability to be easily functionalized, resulting in tunable thermal properties. For example, Feng et al. studied PEG-based PCMs supported by MWCNTs with different functional groups, referred to herein as PEG/MWCNT-X with X as a substitute for carboxyl (-COOH), hydroxyl (-OH), or amine (-NH_2) groups.[96] The DSC curves for hydroxylated, carboxylated, and aminated PEG/MWCNTs exhibited lower phase change temperatures and enthalpies than unfunctionalized PEG/MWCNTs, suggesting that the modified MWCNTs had stronger interaction with PEG than unmodified MWCNTs. The phase change behavior for PEG/MWCNT-X PCMs was mostly affected by hydrogen bonding, capillary forces, and surface adsorption. Thus, the carboxyl functional group had the greatest influence on the phase change behavior of the composite due to strong hydrogen bonding. PEG/MWCNT-COOH exhibited a T_m and T_c of 60.1 °C and 33.7 °C (a 3 °C and 9.9 °C respective reductions from unfunctionalized PEG/MWCNT) and about 40 J/g lower ΔH_m and ΔH_c than PEG/MWCNTs). BET analysis showed that unfunctionalized PEG/MWCNT had a greater surface area at 224 m^2/g and average pore size at 33.2 nm than the MWCNT-X composites suggesting more adsorption sites when unfunctionalized, resulting in increased PEG encapsulation but higher risk of leakage.

CNTs enhance photoconversion efficiency. Sun et al. found that MWCNTs increased the light absorption of the PCM two-fold with a photothermal efficiency of 86.4%.[94] In similar work, Yan et al. also used carboxyl functionalized MWCNTs to increase the photothermal conversion efficiency of PEG because MWCNT-COOH has a high light absorption capacity.[97] Increasing MWCNT content from 0.37 wt% to 3 wt% increased photothermal conversion efficiency from 85.8% to 88.3%, owing to improved absorption in the visible light spectrum. In contrast, PEG alone did not have significant absorption in the visible light spectrum and could not melt under simulated 1000 W/m^2 irradiation. Hence, even small amounts of MWCNTs can significantly improve solar-to-heat energy conversion.

Overall, doping PEG with CNTs is a promising approach to increasing the thermal conductivity, photothermal conversion efficiency, and electrothermal conversion efficiency of PEG, even with small amounts of CNTs. The performance of PEG/CNT composite PCMs is summarized in Table 2.

Table 2. Summary of the phase change properties of PEG/CNT composite PCMs.

Composite	Composition	Synthesis	T_m (°C)	T_c (°C)	ΔH_m (J/g)	ΔH_c (J/g)	Thermal conductivity (W/m·K)	Ref.
PEG/SWCNT	90 wt% PEG	Melting-mixing impregnation	56.7	42.7	179.4	163.6	3.43	[63]
PEG/DCNT	60 wt% PEG	Vacuum impregnation	56.8	42.7	107.4	98.1	1.52	[93]
PEG/$CaCl_2$/MWCNT	80 wt% PEG/$CaCl_2$	Ligand substitution & liquid mixing	49.7	33.2	89.8	–	0.90	[94]
PEG/MWCNT	98 wt% PEG	Self cross-linking reaction	64.5	43.6	120.9	119.4	0.44	[95]
PEG/MWCNT-COOH	90 wt% PEG	Blending & impregnation	60.1	33.7	110.0	103.4	–	[96]
PEG/CNT/diatomite	51 wt% PEG	Vacuum impregnation	8.8	7.21	62.9	69.5	0.29	[98]
PEG/PU/CNT	95 wt% PEG	Ultrasonic dispersion impregnation	42.0	33.9	85.9	75.9	0.50	[99]
PEG/PU/CNT	95 wt% PEG/PU	Ultrasonic dispersion impregnation	61.7	42.2	109.5	105	0.51	[100]
PEG/CNT/Cr-MIL-101-NH_2	70 wt% PEG	Impregnation	52.4	27.4	96.2	90.1	0.46	[101]
PEG/cellulose/MWCNT	80 wt% PEG	Chemical grafting	51.8	32.9	134.7	131.2	0.60	[102]

Acronyms: SWCNT (single-walled carbon nanotubes); DCNT (diatomite/CNT); MWCNT-COOH (multi-walled carbon nanotubes with carboxyl functionalization); CNT (carbon nanotubes); PU (polyurethane); Cr-MIL-101 (chromium terephthalate metal–organic framework).

3.2.2. 2D graphene

2D graphene and its derivatives are primarily used as dopants with extremely high thermal conductivities (around 4000 W/mK).[61] Qian et al. found that 2D graphene nanoplatelets (GNPs) improved the thermal properties of PEG more than 1D SWCNTs (Sec. 3.2.1).[62] 8 wt% SWCNT or 4 wt% GNP were added to PEG to yield PEG/SWCNT and PEG/GNP respectively. The PEG/GNP composite had a 14% higher thermal conductivity than PEG/SWCNT despite using a lower concentration of GNPs than SWCNTs. DSC analysis showed that PEG/GNPs had a lower T_m and higher T_c than PEG/SWCNTs. PEG/GNP also had a 4% and 5% higher ΔH_m and ΔH_c, respectively, than PEG/SWCNT. This was likely because of the slightly greater PEG content in PEG/GNP than PEG/SWCNT. PEG/GNP also had a higher photothermal conversion efficiency than PEG/SWCNT. PEG/GNP showed superior performance than PEG/SWCNT primarily due to the GNPs being 2D compared to the 1D morphology of SWCNTs.

The tradeoff between thermal conductivity and latent heat is partially due to disruptions to PEG crystallinity and decreased PEG content. Marcos et al. found that a 0.5 wt% loading of GNPs increased the thermal conductivity of PEG by 23% while reducing ΔH_m by 7%.[103] The reduction in ΔH_m was partly due to the decrease in PEG content and also partly due to the nanoconfinement of PEG within the layered structure of the GNPs, which decreased PEG crystallinity from 90 to 83%. Thermal diffusivity and viscosity for the GNP composite PCM were 21% and 24% higher, respectively, than pristine PEG at 60 °C. The increased viscosity could reduce PEG leakage during phase change. Thus, an optimization of GNP concentration must be performed to achieve high thermal conductivity and latent heat.

Graphene functionalization enables fine-tuning of the thermal and physical properties of the PEG composite. For example, Selvaraj et al. found that the acidic functionalization of graphene increased its dispersibility in PEG by increasing its zeta potential.[104] PEG modified with acid-functionalized graphene exhibited 72% increased thermal conductivity than pristine PEG, owing to the higher thermal conductivity of modified graphene. Yuan et al. used computational models to show that functionalized graphene would exhibit significantly different phase transition temperatures, latent heats, and stress resistance.[105] Pristine graphene-modified PEG had lower phase change temperatures than PEG, but functionalized graphene-modified PEG had higher phase transition temperatures. The latent heat of graphene was affected by functionalization, with carboxyl > amino > hydroxyl > ethyl > pristine graphene. The cause of the trend was suggested to be the polarity of the functional groups, as more polar carboxyl could interact with PEG through strong intermolecular hydrogen bonding, increasing the energy required to phase change. However, the physical strength of the material decreased, with both Young's modulus and shear modulus being lowest for carboxyl-functionalized graphene and highest for unfunctionalized graphene. This was likely due to the decrease in conjugated double bonds with functionalization.

Graphene oxide (GO), an oxygen-functionalized derivative of graphene, can endow form stability and increase thermal conductivity. For example, Qi et al. prepared PEG-based PCM composites, which used GO nanosheets as a shape-stabilizing host and GNPs as conductive fillers to enhance the thermal and electrical conductivities of the composites.[106] Unlike 3D porous carbons (Sec. 3.2.3), 2D graphene does not generally endow form stability because it lacks a continuous porous structure that can entrap the PEG chains. Rather, graphene-based dopants rely on strong interactions between its functional groups and PEG to immobilize the PEG chains. The PEG/GO/GNP PCM composite showed the best comprehensive performance with 2 wt% GO and 4 wt% GNP having 98% of the latent heat of pure PEG, higher thermal conductivity at 1.72 W/mK (490% higher than pure PEG), and better electrical conductivity at 2.5 S/m. The composite also showed shape stability due to hydrogen bonding between GO and PEG and strong network bonds between GO and the wrinkled surface texture. PEG/GO/GNP PCM composite also showed excellent thermal reliability with negligible change in phase change temperature or enthalpy after 150 thermal cycles.

A limitation of graphene as a shape stabilizing host is its lack of a continuous porous structure that can entrap PEG. To overcome this limitation, a porous 3D network can be fabricated by cross-linking the nanosheets. Qiu et al. cross-linked 2D GO nanosheets to form a 3D porous network to increase encapsulation efficiency.[107] The 3D GO matrix maximized PEG filling and increased the void ratio of the 3D graphene oxide network. DSC showed that the PEG/GO composites had similar phase change temperatures, and a 0.5 wt% GO loading exhibited the highest ΔH_m and ΔH_c at 218.9 J/g and 213.2 J/g, respectively, only 2% lower than pure PEG. The thermal conductivity of the composites improved from that of pure PEG with increasing GO content, showing a minimum enhancement of 63.0% with 0.5% GO and a maximum enhancement of 888% with 6.0 wt% GO. The prepared PEG/GO composites showed high energy density and high thermal conductivity, highlighting their potential for applications in TES systems. Graphene aerogel has been used to achieve 99 wt% PEG encapsulation, one of the highest ever reported.[108] The lignin-modified graphene aerogel was impregnated with PEG *via* vacuum impregnation. The resulting composite PCM had nearly 3-times-higher thermal conductivity and also increased photothermal conversion efficiency.

One of the main benefits to graphene-based modification is improved photothermal energy conversion efficiency. Li et al. prepared composite PCMs based on PEG and GO as photothermal conversion materials.[109] The UV-vis spectra showed that the GO/PEG composite absorbed visible light in the range of 300–800 nm. Irradiation testing with a fluorescent lamp showed that GO/PEG had increasing photothermal conversion efficiencies, increasing from 75% to 87% with increasing GO content from 5 to 15 wt%. Furthermore, infrared thermography showed that while heating, the PEG/GO composites exhibited color change faster than pure PEG, and the color changed faster with increasing GO content, suggesting rapid heat transfer. GNPs also improve the photothermal energy conversion of PEG-based PCMs. He et al. fabricated PEG/GNP composites that exhibited improved visible light absorption for enhanced photothermal performance.[65] The thermal conductivity of the composite with 2 wt% GNPs improved by 146% than pristine graphene with a tradeoff of 6.3% decrease in PEG's latent heat, increasing the rate at which solar irradiation can spread below the surface of the PCM. Thus, PEG/GNP composite PCMs are promising for application that relies on solar heating.

Graphene is an excellent nano-additive for PEG due to its high thermal conductivity, functionalizability, facile and green synthesis, diverse morphology, and high

Table 3. Summary of the phase change properties of PEG/graphene composite PCMs.

Composite	Composition	Synthesis	T_m (°C)	T_c (°C)	ΔH_m (J/g)	ΔH_c (J/g)	Thermal conductivity (W/m·K)	Ref.
PEG/GNP	96 wt% PEG	Vacuum Impregnation	56.5	43.7	169.3	157.6	3.11	[62]
PEG/GNP	98 wt% PEG	Temperature-assisted solution blending	56.1	37.1	170.6	151.3	0.78	[65]
PEG/GNP	99.5 wt% PEG	Vortex mixing and ultrasonication	5.35	3.25	104.4	–	0.19	[103]
PEG/GO/GNP	94 wt% PEG	Physical blending	58.3	44.9	167.4	153.8	1.72	[106]
PEG/GO	94 wt% PEG	In-situ filling	60.5	40.5	218.9	213.2	0.39	[107]
PEG/LGA	99 wt% PEG	Vacuum filtration	61.0	41.9	168.7	165.12	0.38	[108]
PEG/GO	85 wt% PEG	Ultrasound-assisted physical blending	42.4	36.1	87.4	78.7	–	[109]

Acronyms: GNP (graphene nanoplateletes); GO (graphene oxide); LGA (lignin-modified graphene aerogel).

compatibility with PEG. Just adding a small amount of graphene can significantly improve the thermal conduction and photothermal conversion efficiencies of PEG. The performance of PEG/graphene composite PCMs is summarized in Table 3.

3.2.3. 3D porous carbons

3D porous carbons are promising hosts for PEG because they can be easily synthesized from renewable, biological sources, such as foods,[70] natural fibers,[110,111] lignin,[112] or wood-derived carbons.[113,114] These biologically derived porous carbons (BPCs) greatly enhance thermal conductivity. For example, Zhang et al. found a 2-fold increase in thermal conductivity by adding a carbonized starch filler material to PEG.[115] DSC results showed that increasing carbon content did not affect the phase transition temperatures but decreased ΔH_m due to the low encapsulation efficiency (62 wt% PEG). Additionally, the supercooling extent in the composites decreased by as much as 3.67 °C.

Unlike carbon-based dopants like 1D CNTs (Sec. 3.2.1) and 2D graphene (Sec. 3.2.2), hierarchical porous carbons can encapsulate PEG to endow form stability. Figure 5(a) depicts the standard vacuum impregnation method of fabricating form-stable PEG/porous carbon composite PCMs.[70] The carbonization temperature and carbon source determine the crystal structure and thermal conductivity of biologically derived porous carbons (BPCs). For example, Zhao et al. found that higher carbonization temperatures yielded BPCs with a higher degree of graphitization, resulting in higher thermal conductivity.[70] The carbonization temperature was optimized at 1300 °C, whereas carbonization temperatures above 1300 °C destroyed the BPC's porous structure. The

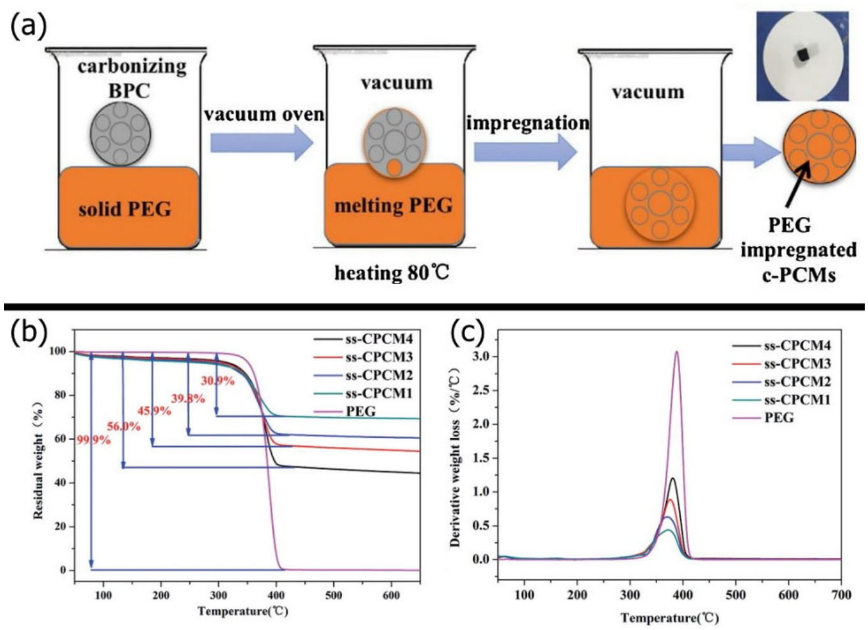

Figure 5. (a) Schematic for the preparation of the PEG/BPC composite *via* a standard vacuum impregnation method.[70] (Reproduced with permission from[70], Copyright Elsevier, 2018). (b) TGA and (c) DTG of ss-CPCM with 25, 33, 40, and 50 wt% PEG labeled ss-CPCM 1 to 4, respectively.[125] (Reproduced with permission[125], Copyright Royal Society of Chemistry, 2016).

PEG/radish BPC composite exhibited the same T_m and T_c as PEG/potato BPC, but the radish-derived composite had a slightly higher ΔH_m and ΔH_c of 159.7 J/g and 155.6 J/g, respectively, because the radish BPC had a higher PEG encapsulation of 89.77%. However, the potato composite had a thermal conductivity of 4.489 W/mK, 157% higher thermal conductivity than the radish composite, and nearly 10-times-higher than pure PEG.

In addition to the host material's porosity, the particle size of the carbon material affects PEG-based PCM performance. Wen et al. prepared two form-stable composite PCMs with bone char (BC) of two different sizes as a porous host.[116] BC was ground from carbonized cattle bones and sieved into two sizes: 0.8–1 mm (larger BC called PEG/BC-L) and 0.25–0.8 mm (smaller BC called PEG/BC-S). BET analysis showed that BC-L had a larger pore size and smaller surface area of 13.5 nm and 112.6 m^2/g, respectively, than BC-S particles (12.2 nm and 116.643 m^2/g). Thus, PEG/BC-S had a higher PEG encapsulation of 44% compared to 39% for the large-BC composite, resulting in a 14% higher ΔH_m with similar phase transition temperatures. Moreover, PEG/BC-S exhibited higher thermal degradation than PEG/BC-L. Hence, the carbon material's particle sizes should be optimized for maximum PEG encapsulation.

Expanded graphite (EG) is another promising porous carbon for its high encapsulation efficiency and thermal conductivity. In a recent study, Bai et al. found that 1 wt% of EG increased the thermal conductivity of PEG/VO_2 by 94% and latent heat by 59%.[117] The EG also increased the maximum PEG encapsulation without leakage from 39 wt% for PEG/VO_2 to 74 w% for PEG/VO_2/EG, owing to the high surface area and porosity of EG. To enhance the performance of PEG/EG, Zhang et al. used a titanate coupling agent (KR-38S) to bridge the gaps between EG and PEG, forming a composite referred to as m-EP.[118] The m-EP composite PCM had a 6% lower thermal conductivity than without the organic coupling agent, but the higher thermal conductivity of m-EP was still 10-times-higher than pristine PEG. The ΔH_m and ΔH_c for the m-EP composite PCM were 10% higher than that of the PEG/EG composites because the titanate coupling agent increased PEG adsorption and prevented leakage during phase transition. Additionally, the m-EP composites decreased supercooling by 3 °C more than the PEG/EG composites. The titanate-modified PEG/EG had a 20 °C higher initial decomposition temperature than PEG/EG, with decomposition starting at temperatures above 300 °C.

Because the pore size of EG significantly affects its ability to host PEG, the pore size of EG must be finely tuned to maximize PEG encapsulation. One method is to synthesize EG using templates. For example, Zhu et al. used a salt template to yield an EG/rubber composite matrix.[119] Higher NaCl content yielded smaller EG/rubber particles, smaller pore sizes, and more interconnected channels. Thus, EG/rubber synthesized with 85 wt% NaCl had a 267% higher PEG encapsulation efficiency than EG/rubber synthesized with 75 wt% NaCl. As a direct result of having higher PEG content, the PEG composite synthesized with 85 wt% NaCl exhibited the highest ΔH_m and ΔH_c of 144.1 and 140.4 J/g, 345% higher than PEG/EG/rubber synthesized with 75 wt% NaCl.

In addition to improving thermal conductivity, most 3D porous carbons increase the photothermal conversion efficiency of PEG-based PCMs. Since PEG reflects most sunlight,[82] it is critical to modify PEG with solar-absorptive porous carbons, especially for solar TES applications (Sec. 4.5). Zhao et al. found that adding 0.5 wt% carbon black increased the solar heating rate.[120] After 300 s of solar irradiation, the carbon black-doped PEG had an average final temperature of 75 °C, whereas pristine PEG was 54.1 °C. Lin et al. found that PEG/carbon foam composite exhibited a high photothermal energy conversion efficiency of 74.4%.[121] This resulted in a 34 °C higher surface temperature for PEG/carbon foam than PEG after 650 s of irradiation. The PEG/carbon foam composite with 10 wt% carbon foam also had a 224% higher thermal conductivity than pristine PEG, which helped distribute the heat to the bulk PCM. Wu et al. found that compositing PEG-$CaCl_2$ with carbon felt increased thermal conductivity by 985% in-plane and 571% through-plane, which greatly improved the heat spread from the surface of the PCM to rest of the system.[122] BPCs can also improve photothermal conversion efficiency. Liu et al. impregnated biochar derived from corn straw with PEG to increase the photothermal efficiency of PEG from 45% without biochar to 69% with biochar under infrared irradiation.[123] The thermal conductivity also increased by 32%, decreasing the phase transition time by 3 times.

An interesting property of PEG/porous carbon composites is their electrothermal conversion efficiency. Sun et al. found that a PEG-$CaCl_2$/EG PCM exhibited a high electrothermal conversion efficiency of 86.9% at 5 V.[124] When a voltage was applied to the composite PCM, the electrons in the EG moved rapidly and generated heat when the carriers were absorbed or scattered. The high electrical conductivity of EG enabled rapid carrier movement and also enabled stable electrothermal conversion, with a less than 2% decrease in electrothermal conversion efficiency after 50 electro-heating cycles.

Impregnating porous carbon hosts can also improve thermal stability and endow anti-flammable properties. Tan et al. found that a solid-state PEG/carbon composite (ss-CPCM) with 25 wt% PEG showed excellent thermal stability at the cost of energy storage capacity (74% lower than pristine PEG).[125] The TGA results in Figure 5(b) shows that ss-CPCM1 (1:4 ratio of PEG to porous carbon) had the quickest weight loss because the high thermal conductivity caused the solvent to evaporate rapidly. Differential thermogravimetry (DTG) analysis in Figure 5(c) indicates that carbon-modified PEG had a much lower rate of PEG decay, with higher carbon content lowering the degradation rate. This occurred for two reasons. First, PEG was adsorbed by the porous carbon and stayed bound to the carbon host. Second, the carbon skeleton acted as an anti-flammable barrier around the encapsulated PEG, reducing the effect of high-temperature heating from the surface.

Conclusively, 3D porous carbon modifiers provide thermally conductive matrices for PEG insertion, improving heat transfer efficiency and reducing leakage. Porous carbon matrices may be derived from organic sources or synthesized in a laboratory to have small pore sizes and high porosity, optimizing for high PEG encapsulation and thermal conductivity. Carbon skeletons increase the photothermal and electrothermal conversion efficiencies, which are important traits for solar and other TES applications. The performance of PEG/3D porous carbon composite PCMs is summarized in Table 4.

3.3. PEG/silica composite

Silica is often used as support materials to reduce leakage and improve the thermal stability of PEG-based PCMs, owing to their high adsorption capacity, thermal stability,

Table 4. Summary of the phase change properties of PEG/carbon composite PCMs.

Composite	Composition	Synthesis	T_m (°C)	T_c (°C)	ΔH_m (J/g)	ΔH_c (J/g)	Thermal conductivity (W/m·K)	Ref.
PEG/potato-carbon	85 wt% PEG	Vacuum impregnation	56.5	37.9	158.8	155.4	4.5	[70]
PEG/carbon/MEV	56 wt% PEG	Impregnation & in-situ carbonization	58.8	41.0	104.4	90.6	0.60	[115]
PEG/BC	61 wt% PEG	Impregnation	58.5	43.0	62.7	59.9	–	[116]
PEG/VO_2/EG	74 wt% PEG	Vacuum impregnation	50.2	34.7	104.3	95.1	0.31	[117]
PEG/EG	7/1	Vacuum adsorption	54.6	41.3	141.6	124.1	3.66	[118]
PEG/EG/rubber	74 wt% PEG	Vacuum impregnation	44.1	37.5	144.1	140.4	1.05	[119]
PEG/carbon foam	74 wt% PEG	Vacuum impregnation	52.2	43.9	156.3	147.4	0.68	[121]
PEG/biochar	50 wt% PEG	Vacuum impregnation	55.7	34.9	100.2	95.1	0.43	[123]
PEG/carbon	50 wt% PEG	Ultrasound Assisted Mixing	56.7	38.9	91.8	81.8	–	[125]
PEG/EG	7/1	Physical blending & vacuum impregnation	48.6	38.9	140.1	138.6	0.24	[126]
PEG/carbon	90 wt% PEG	Vacuum impregnation	61.6	19.9	149.0	138.1	–	[127]
PEG/carbon	90 wt% PEG/carbon	Physical impregnation	49.2	29.8	88.2	79.9	0.31	[128]
PEG/carbon	92.5 wt% PEG	Physical blending and impregnation	60.0	25.0	162	158.2	0.42	[129]

Acronyms: MEV (modified expanded vermiculite); BC (bone char); EG (expanded graphite).

and functionalizability.[130,131] The silica-based hosts may be from nature like diatomite (Dt) or mica,[132,133] or synthesized in a lab, like mesoporous silica, aerogels, and glass.[134,135] For example, Yang et al. prepared form-stable PCM composites based on PEG and hollow glass microspheres (HGS) as a support material.[135] SEM showed that the composite had a rough stacked morphology of broken HGS spheres, which worked to increase the surface area of the composite. PEG was well distributed on the surface and in the pores of the support material. The maximum loading of PEG was up to 75 wt% without leakage, which is lower than for most porous silica hosts. The low PEG content resulted in 24% decreased latent heat than pure PEG. However, the composite showed good thermal stability, with negligible change in latent heat after 100 thermal cycles. The composite also boasted excellent thermal stability, with serious degradation only at temperatures greater than 300 °C, owing to the high thermal stability of HGS.

Highly porous silica materials generally have higher PEG encapsulation efficiencies than polymer (Sec. 3.1), porous carbons (Sec. 3.2.3), and other host materials (Sec. 3.4), resulting in large latent heats. Li et al. prepared a shape-stabilized composite PCM based on PEG PCM and SiO_2 as a support matrix (PEG/SiO_2) using a novel synthesis method based on a common process used for mesoporous silica.[69] The maximum encapsulation of PEG without leakage was 97.3 wt%, one of the highest reported for form-stable PEG PCMs. Hence, DSC analysis showed that the PEG/SiO_2 composite had ΔH_m and ΔH_c only 8% and 4% lower than pristine PEG, respectively. The optimized composite showed great thermal reliability through 100 heating-cooling cycles, with negligible change in phase change temperature and latent heat.

Zeolites, porous aluminosilicate materials, have gained interest as support materials for PEG due to their large hosting capacity and thermal conductivity. Li et al. prepared form-stable composite PCMs using PEG PCM and Zeolite Socony Mobil–5 (ZSM-5) as a porous support material.[136] The PEG/ZSM-5 composites were prepared *via* physical mixing followed by vacuum impregnation. During vacuum impregnation, PEG was absorbed into the ZSM-5 pores; and when pore space was exceeded, PEG was adsorbed onto the ZSM-5 surface *via* surface tension and capillary effects. However, the maximum encapsulation of PEG without leakage was only 50 wt%. The PEG content adhered to the surface of ZSM-5 can crystalize more freely, which results in larger enthalpy than in low PEG content composites. The PEG/ZSM-5 composites also showed reduced supercooling and 200% higher thermal conductivity than pristine PEG.

Mesoporous silica materials are especially popular as hosts for PEG for their high adsorption capacity, controllable porosity, and high thermal conductivity.[137,138] Work by Feng et al. found that the functional groups present in mesoporous silica significantly affected the thermal performance of PEG as a PCM.[139] A mesoporous silica material, MCM-41, was grafted to prepare amine-terminated silica. DSC curves showed that prior to modification (PEG/MCM-41-OH) did not show an endothermic peak, indicating that PEG phase change did not occur with hydroxyl groups despite having a maximum PEG encapsulation of 70 wt%. However, after modification with amino groups (PEG/MCM-41-NH_2), a phase change occurred even with only 30 wt% PEG encapsulation. The phase change enthalpy increased with increasing PEG content almost 5 fold to 58.76 J/g when doubling PEG content from 30 to 60 wt%. Through molecular dynamics methods, it was determined that strong interactions between PEG and the hydroxyl groups in MCM-41-OH resulted in PEG being excessively adsorbed onto the surface of MCM-41-OH, hindering PEG phase change. However, after modification with amino groups, MCM-41-NH_2 and PEG showed reduced intermolecular interactions, allowing PEG to change phase. The composites showed thermal reliability through 200 thermal cycles with negligible change in phase change temperature and enthalpy. Wang et al. similarly found that reducing the intermolecular interactions between PEG and the silica support can increase latent heat.[140] PEG with surface-modified mesoporous silica, SBA-15, was modified with hydroxyl, amine, or methyl groups terminated group (referred to herein as PEG/HO-SBA-15-OH, PEG/NH_2-SBA-15-NH_2, or PEG/NH_2-SBA-15-CH_3, respectively). DSC analysis showed that PEG/HO-SBA-15-OH showed no fusion or solidification peaks. In contrast, amino-functionalized PEG/NH_2-SBA-15-NH_2 showed similar crystallization temperature to pure PEG, likely due to significant PEG leakage. Amino modifications of SBA-15 reduced the hydrogen bonding interaction of PEG and support material, resulting in decreased PEG adsorption and adjusted PEG chain configuration from a long chain to a loop structure. Both of the aforementioned factors improved the stretching and crystallization of PEG's chains, which improved the thermal properties of the composite. The introduction of the methyl terminal groups to amino-functionalized SBA helped to reduce PEG leakage due to the different polarities of PEG and methyl groups. Thus, the crystallization behavior and, thereby, thermal properties of PEG/silica PCMs may be controlled by tuning surface functionalization.

Silica-based hosts for PEG are highly compatible with various dopants that can improve the thermal performance of PEG. Chen et al. prepared shape-stable PCM composites based on PEG and silica fume (SF) with dopamine (Dop) modification as a supporting material (PEG/Dop-SF).[141] The latent heat of PEG/Dop-SF was higher by 9.8% than dopamine-less PEG/SF due to the dopamine converting some hydroxyl groups in the SF into imino groups. The imino groups had weaker interactions with PEG, allowing for better crystallization and, consequently, higher latent heat. Moreover, the modification of SF with Dop improved the crystallization ability of the composite PCM and provided additional benefits to the composite, such as nontoxicity and safer operating conditions. During Dop-SF modification, the o-dihydroxybenzene from Dop interacted with the hydroxyl terminated groups on SF, yielding polydopamine (PDA) with free imino groups. The PEG/Dop-SF composite formed through host-guest hydrogen bonding interaction between PEG's hydroxyl groups and Dop-SF's free imino groups, reducing PEG leakage. In another example, Qian et al. prepared shape-stabilized PCMs based on PEG and Ag nanoparticles (AgNPs) dispersed in a diatomite matrix as a supporting material (PEG/DtAg).[142] The maximum loading of PEG in the composite without leakage was 63 wt%, resulting in a 38% lower ΔH_m than pure PEG. However, the supercooling extent of the PEG/DtAg composites improved with a 16.3% reduction than pure PEG and 10% lower supercooling than Ag-less PEG/Dt. The thermal conductivity of the PEG/DtAg (0.82 W/mK) was also 127% higher than PEG/Dt. These studies highlight how organic and metallic dopants are compatible with PEG/silica composites to enhance PCM properties.

Mesoporous silica sheets may use highly porous graphene nanosheets to adopt a highly porous structure. For example, Zhang et al. prepared PEG-based PCM composites absorbed into mesoporous graphene-based silica sheets (GS) to reduce PCM leakage and improve PEG conductivity without phase separation.[143] GS were synthesized *via* a sol-gel and templating method with a GO template and tetraethyl orthosilicate, yielding silica sheets immobilized on graphene sheets. GS had a 5-times-larger BET surface area than the GO template, owing to an increase in pore size. The larger pore volume enabled high encapsulation up to 80 wt%, but the PEG/GS composite could only hold 70 wt% PEG without leakage. The measured enthalpies were lower than theoretically predicted based on the PEG loading, indicating that movement of the PEG chains was restricted by the GS's steric and drag effects at the molecular level.

An interesting morphology of silica-based hosts is nanowires, owing to their flexibility and encapsulation efficiencies. Zhang et al. fabricated a PEG-based PCM composite with Si_3N_4 nanowires using an ultrasonic cell disruptor method.[64] SEM showed that PEG is well absorbed in the fluffy Si_3N_4 nanowire support matrix, with a high PEG encapsulation of 90 wt% and minimal leakage. The ΔH_m and ΔH_c were only 17% lower than pristine PEG due to the high encapsulation efficiencies. Using ultrasonic cell disruptor methods for PCM synthesis is advantageous because it takes significantly less preparation time (0.5 h) than standard ultrasonic methods (25 h). The composites were thermally stable even after 100 thermal cycles. The thermal conductivity of the PEG/Si_3N_4 composites was a maximum of 0.362 W/mK, an 89% improvement from pristine PEG. More work should be done in fabricating silica-based nanowires or nanofibers as hosts for PCMs with lowered phase transition temperatures for their potential applications in smart textiles.

Silica is an excellent support material for PEG, owing to its high encapsulation capacity, diverse morphology, low cost, and nontoxicity. Their high thermal stabilities make PEG-based PCMs more viable for high-temperature applications. The performance of PEG/silica composites is summarized in Table 5. The effects of their functionalization and modification should be investigated more, as silica-based materials are easily customizable with the potential to endow additional properties like photothermal conversion efficiency and flame retardancy.

3.4. PEG/other composite

This section discusses recent studies involving various uncommon but promising host materials and dopants used to enhance PEG-based PCMs. Some of the materials covered in this review include LDHs, metal oxides, metal foams, MOFs, metal nanoparticles, boron nitride, graphitic carbon nitrides, and MXenes. The advantages of using the specific modifiers are analyzed, emphasizing the properties are important for improving the PCM performance of PEG.

LDHs have gained much attention for their versatile hosting ability and easily tunable functionalization.[145] Zhu et al. prepared shape-stabilized PCM composite consisting of PEG hosted in surface-modified LDHs.[146] The surfaces of the LDHs were modified with 3-aminopropyl triethoxysilane (KH), which enabled the carrier to maintain a solid form during PEG phase transition. The PEG/KH-LDH composites were synthesized by solution impregnation with varying ratios of PEG; however, it was found that PEG had a maximum encapsulation of only 55% without leakage, lower than most polymer,

Table 5. Summary of the phase change properties of PEG/silica composite PCMs.

Composite	Composition	Synthesis	T_m (°C)	T_c (°C)	ΔH_m (J/g)	ΔH_c (J/g)	Thermal conductivity (W/m·K)	Ref.
PEG/SiO_2	97 wt% PEG	Mesoporous silica method	60.4	37.9	164.9	160.1	–	[69]
PEG/Novolac/Cloisite 15 A	85/12/3	In-situ sol-gel polymerization without solvent	52.0	24.9	188.8	178.5	–	[72]
PEG/mica	45 wt% PEG	Vacuum impregnation	46.0	–	77.8	77.7	0.37	[132]
PEG/HGS	75 wt% PEG	Impregnation	58.9	45.3	124.8	123.7	–	[135]
PEG/ZSM-5	60 wt% PEG	Physical blending & vacuum impregnation	59.9	46.2	115.6	96.8	0.54	[136]
PEG/MCM-41-NH_2	60 wt% PEG	Solution impregnation	50.8	–	58.8	–	–	[139]
PEG/NH_2-SBA-15-CH_3	70 wt% PEG	Blending & impregnation	52.0	30.0	88.2	82.2	–	[140]
PEG/Dop-SF	70 wt% PEG	Vacuum impregnation	53.0	44.8	73.8	69.1	–	[141]
PEG/DtAg	63 wt% PEG	Vacuum impregnation	59.5	39.5	111.3	102.4	0.82	[142]
PEG/graphene-silica	80 wt% PEG	Mixing Adsorption	49.2	30.0	136.3	133.8	–	[143]
PEG/SiO_2	80 wt% PEG	Temperature-assisted sol-gel	28	26.6	113.6	115.4	0.28	[144]

Acronyms: HGS (hollow glass microspheres); ZSM (zeolite socony mobil–5); MCM-41 (Mobil Composition of Matter No. 41); SBA-15 (Santa Barbara Amorphous-15; Dop-SF (dopamine-silica fume); DtAg (diatomite doped with Ag nanoparticles).

carbon, and silica hosts. While LDHs are theoretically good candidates because of their highly porous, lamellar structure, the stability of PEG encapsulation must first be improved.

Instead of impregnating a lamellar structure, such as LDHs, PEG has been encased in core/shell morphologies to reduce PEG leakage. Deng et al. synthesized a flower-like nanostructured TiO_2 (FLN-TiO_2) to encapsulate PEG, reducing PCM leakage during phase transition and providing shape stability to the composite for longer-term thermal cycles.[147] Figure 6(a) shows a schematic representation of the PEG/FLN-TiO_2 fabrication process. The initially synthesized composite PCM had excess PEG, which leaked out excessively onto filter paper at 80 °C for 12 h (Figure 6b and 6c). Figure 6(d) shows the form-stable PEG after all the excess PEG was removed. SEM showed that PEG was well encapsulated in a spherical FLN-TiO_2 shell and distributed throughout the pores and surfaces of FLN-TiO_2, resulting from capillary forces and surface tension effects. The PEG/FLN-TiO_2 possessed form stability with minimal leakage and at a maximum encapsulation of 50.2 wt%. PEG/FLN-TiO_2 exhibited weak interfacial physical interactions, which reduced the phase change temperature of the composite. PEG/FLN-TiO_2 showed thermal stability, up to 200 melting/freezing cycles, suggesting that from-stable FLN-TiO_2 improved the thermal reliability of the composite. Such core/shell PEG PCMs would be promising for their versatility as filler materials in cement, asphalt, and HTFs. While most PEG-based core/shell morphologies have PEG at the core, PCMs may have higher TES capacities when inverted to have PEG shells. For example, Wang et al. fabricated a stearic acid/PEG core/shell microcapsule PCM that exhibited a high ΔH_m and ΔH_c of 115.3 and 113.4 J/g, respectively.[148] While stearic acid/PEG core/shell was not contrasted to inverted PEG/stearic acid core/shell, such findings could be important in shifting how to design PEG composite PCM capsules.

Metal foams are another common support material for PEG, owing to their high thermal conductivities, low supercooling, and excellent thermal resistance. For example, Ren et al. impregnated nickel foam with PEG, increasing the thermal conductivity of the composite PCM by 157% and decreasing supercooling by 35%.[149] To further increase the porosity of the nickel foam (Figure 6e), the foam was modified to have flower-like nanoarrays on its surface (Figure 6f and 6g), which acted as nucleation sites for PEG crystallization, reducing supercooling. Figure 6(h) shows a magnified SEM of the porous, flower-like $Ni(OH)_2$. The high porosity but small pore sizes enabled high encapsulation of 80 wt% PEG, indicated by a dense coating of PEG over the modified Ni foam surface (Figure 6i and 6j). The inset in Figure 6(i) shows how the resulting PEG/modified Ni foam composite retained its macroporous structure. PCM Similarly, Zhao et al. found that porous TiO_2 foam increased thermal conductivity by 43% and endowed form-stability.[150] While the latent heat decreased by 32% as expected due to decreased PEG content, the higher thermal conductivity decreased the average half crystallization time by 11% and nucleation activation energy by 18%.

Metal-organic materials are also popular as host materials for their high porosity and customizability.[151] For example, Wang et al. prepared a shape-stable PEG PCM composite with an iron-benzenetricarboxylate metal-organic gel using a 3:2 metal-to-ligand ratio, referred to herein as PEG/Fe-MOG-100.[152] The composite was synthesized by an in-situ assembly strategy where PEG is introduced and are aggregated into a gel with

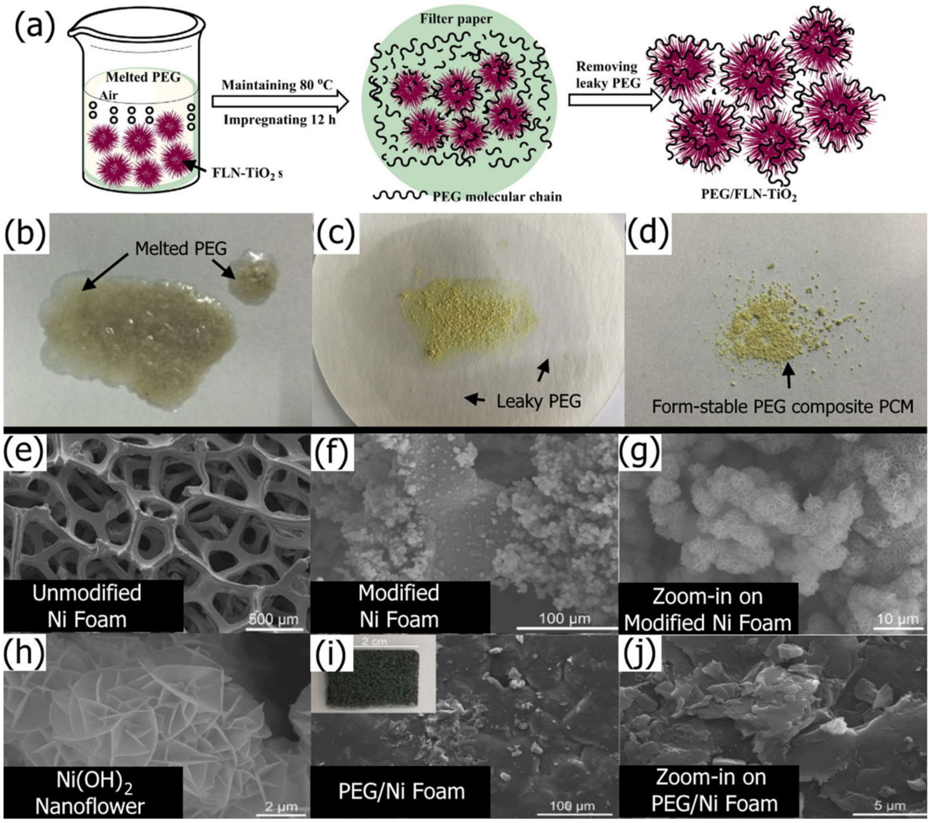

Figure 6. (a) Schematic representation of the synthesis of PEG/FLN-TiO_2. Digital photographs of (b) PEG, (c) PEG/FLN-TiO_2 with liquid leakage trace, and (d) PEG/FLN-TiO_2 with good shape stability after leaky PEG removal.[147] (Adapted with permission from [147], Copyright Elsevier 2017). SEM of (e) unmodified Ni foam and (f) $Ni(OH)_2$ nanoflower-modified Ni foam. (g) Magnified SEM view of $Ni(OH)_2$ nanoflower-modified Ni foam. (h) Magnified SEM view of $Ni(OH)_2$ nanoflower. (i) SEM of PEG/modified Ni-foam composite PCM with a digital photograph insert. (j) Magnified SEM of PEG/modified Ni-foam composite PCM.[149] (Adapted with permission from [149], Copyright Elsevier 2022).

MOG-100-Fe, synthesized *via* a sol-gel method. 92 wt% PEG/Fe-MOG-100 with metal to ligand ratios of 1:4 had a high ΔH_m that was 98% of pristine PEG's latent heat, one of the closest ever reported to the theoretical maximum. Altering the metal-to-ligand ratio and the weight percent of PEG present regulated the thermal properties of the composite. The PEG/Fe-MOG-100 composites showed reduced supercooling when the metal to ligand ratio was equal to or higher, with the greatest reduction seen in 92 wt% PEG/Fe-MOG-100 with a 1:3 metal to ligand ratio, which showed a 15% lower supercooling extent than pure PEG. In more recent work, Wang et al. formed a hydrogel-like PEG composite using a metal-organic network gel.[153] The porous and polar support structure allowed one of the highest encapsulation efficiencies of 98 wt% PEG, resulting in 97% of the ΔH_m of pure PEG. The metal centers in the metal-organic network also slightly increased thermal conductivity by 22%. These metal-organic gels have shown some of the highest PEG encapsulation efficiencies reported and are promising hosts that endow form stability and reduce supercooling.

One of the most promising aspects of MOFs is their easily tunable porosity, which depends on the organic ligands used. Uemura et al. compared the effects of 7 different ligands on encapsulated PEG's thermal properties using various computer simulations and experimentation. PEG was successfully encapsulated by the MOF with an encapsulation efficiency of 45 wt%. By using shorter organic ligands, the pore size of the MOF was decreased, resulting in less PEG encapsulation but reduced leakage. PEG/MOFs with smaller pore sizes had significantly lower phase transition temperatures, as low as −53 °C with styrenedicarboxylate ligands. Moreover, increasing the MW of the confined PEG resulted in decreased phase transition temperatures, contrary to what is typically expected. This was caused by the destabilization of PEG when encapsulated, with larger PEG chains having more significant destabilization when confined, resulting in a lower melting temperature. Thus, controlling the pore sizes may decrease the phase transition temperature of larger MW PEG.

The addition of metal nanoparticles like Ag often results in greater light absorption capability, increasing photothermal conversion efficiency. For example, the PEG doped with Ag nanoparticles and graphene nanosheets (GNS) synthesized by Zhang et al. (Figure 7a) had greater full-band absorption than AG-less PEG/GNS.[154] The Ag-GNS/PEG composites also showed reduced thermal radiation. The improved light absorption and reduced thermal radiation were due to the localized surface plasmon resonance effect of the Ag nanoparticles illustrated in Figure 7(b). The Ag-GNS/PEG composites showed enhanced solar-thermal conversion efficiency of 92.0%, much higher than PEG (70.8%), making the composite PSC suitable for various applications in solar energy harvesting. Similarly, Xiao et al. found that adding Ag nanoparticles to PEG/carbon increased photothermal conversion efficiency from 77% to 92%, owing to the aforementioned local surface plasmon resonance effect.[155]

Conductive nanowires may also be used as filler materials, especially to enhance the thermal conductivity of PEG. For example, Deng et al. prepared a PEG-based shape stabilized PCM composite using PEG-modified silver nanowire (Ag NW) filler and expanded vermiculite (EV) support material.[156] SEM revealed that PEG-coated silver nanowires were well distributed in the pores and surface of EV, owing to surface tension and capillary forces. The maximum encapsulation of PEG possible in the composite while maintaining suitable shape stability was 66.1 wt%. Ag NWs acted as a filler to enhance thermal conductivity to 0.88 W/mK, 11.3 times higher than pristine PEG. However, the dopants reduced the thermal energy storage ability of PEG. The length and diameter of the Ag NW played an important role in the thermal conductivity of the composite, with a length of 5–20 μm and a diameter of 50–100 nm yielding the best results.

Other dopants in the crystalized PEG matrix, such as metal oxides, play a significant role in improving thermal conductivity,[157] viscosity,[158] hysteresis,[159] and other fluid or thermal properties. Inorganic salts, such as ammonium polyphosphate (APP),[160] may play a crucial role in endowing fire resistance. Yin et al. fabricated a PEG/poly (glycerol-itaconic acid)/APP (PEG/PGIA/APP) PCM that suppressed fire spreading by creating inflammable char layers.[160] The PEG composite had an increased limiting oxygen index from 21.6% (without APP) to 28.7% with APP and a 36% decreased pHRR of 413 kW/m^2. The increased limiting oxygen would reduce the likelihood and

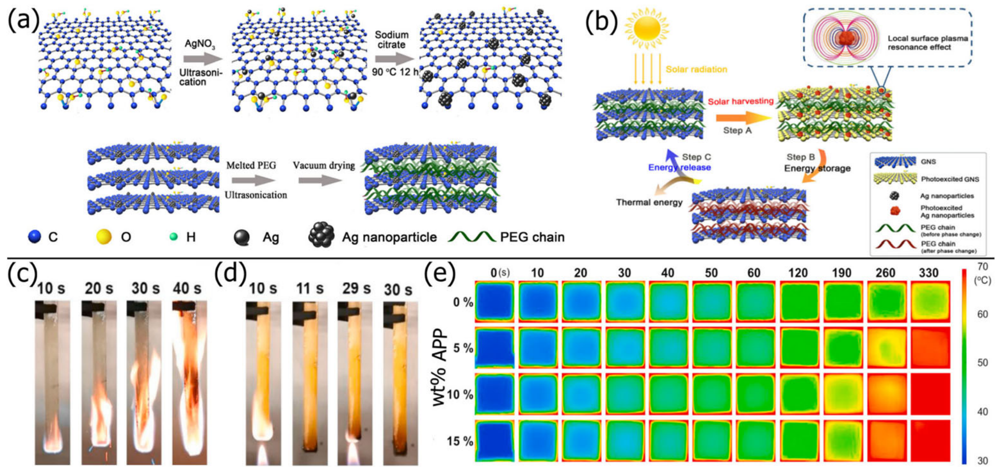

Figure 7. (a) Schematic showing the synthesis process of PEG/Ag-GNS. (b) Schematic for the light-to-thermal energy conversion and storage of the Ag-GNS/PEG composites *via* localized surface plasmon resonance effect.[154] (Reproduced with permission from [154], Copyright Elsevier, 2019). Digital photographs showing the flame resistance of (c) PEG and (d) PEG/PGIA/APP composite PCM. (e) IR Images showing the distribution of thermal energy after 330 s with different APP content.[160] (Modified with permission from [160], Copyright Elsevier, 2022).

duration of the flame, while the decreased pHRR would limit the amount of heat released during a potential fire. Figure 7(c) shows how PEG is quick to catch fire, whereas Figure 7(d) shows the formation of a char layer in the PEG/PGIA/APP composite PCM, which prevents the combustion of PEG. Moreover, the thermal conductivity improved by 36% with 15 wt% APP than pristine PEG, resulting in rapid heat dispersion indicated *via* IR imaging in Figure 7(e). The thermal dispersion also improved flame resistance by spreading out concentrated thermal energy, lowering the chance of combustion.

Boron nitride (BN) is a common ceramic filler for thermally conductive composite materials owing to its high thermal conductivity, small density, low cost, chemical stability, and low electrical conductivity.[161–163] Yang et al. prepared PEG-based PCM with boron nitride (BN).[164] DSC showed that increasing BN content decreased latent heat due to decreased PEG content. However, compared to pristine PEG, supercooling decreased from 23.5 to 21.1 °C, thermal conductivity increased by 336%, and photothermal conversion efficiency increased to 72.7% with 30 wt% of BN. The composite showed thermal reliability through 100 thermal cycles, with less than a 2.8% change in latent heat of fusion and good shape stability. While BN is a promising additive, it is difficult to composite non-polar BN with polar PEG. To improve the interfacial adhesion between PEG and BN, BN has been modified with polar functional groups. Wie et al. functionalized BN with polyvinyl acetate before cross-linking with PEG, resulting in 46% higher thermal conductivity than unfunctionalized PEG/BN.[165]

Graphitic carbon nitrides have also been used as support materials for PEG, owing to its high thermal conductivity, structural stability, and mesoporosity. Feng et al. synthesized form-stable PEG-based PCMs with graphitic carbon nitrides (bulk-C_3N_4) and carbon nitride intercalation compounds (CNIC) as support materials *via* blending and impregnation methods, respectively.[166] SEM showed that PEG/bulk-C_3N_4 has a smooth morphology from PEG covering the irregular surface of bulk-C_3N_4, resulting in a low surface area of 5 m^2/g. In contrast, PEG/CNIC possessed a porous structure with nanorod aggregation, yielding a high surface area of 77 m^2/g. Hence, the maximum encapsulation of PEG without leakage was only 40 wt% for PEG/bulk-C_3N_4 but 60 wt% for PEG/CNIC. The DSC curves for PEG/bulk-C_3N_4 showed no endothermic or exothermic peaks, indicating that bulk-C_3N_4 acted as an impurity to PEG crystallization, impairing the composite's phase change ability. In contrast, PEG/CNIC showed distinct endothermic and exothermic peaks that were 35% lower than that of pristine PEG, owing to the low PEG loading and hindered PEG mobility. The differences in thermal performance between PEG with bulk-C_3N_4 and CNIC support matrices were attributed to the significantly more porous morphology of CNIC that allowed PEG crystallization, highlighting the importance of the carbon support material's structure.

MXenes are promising 2D materials that have been used to improve the photothermal conversion efficiency of PEG-based PCMs, owing to MXene's high thermal and electronic conductivities, abundant active sites, and local surface plasmon resonance effect.[80] In a recent study, Fang et al. increased the photothermal conversion efficiency by doping MXenes.[167] While PEG has poor photothermal conversion efficiency due to its reflectivity, metal carbonitride MXenes have excellent solar absorption, making it a promising additive to solar-heated PEG-based PCMs. In work by Fang et al., PEG was impregnated into potato-derived porous carbon and doped with MXene.[167] The porous carbon host enabled form stability and increased thermal conductivity but had poor PEG encapsulation of 61 wt%. Adding porous MXenes increased PEG encapsulation to 82.1 wt% because of the abundant hydrogen bonding between PEG and the MXene surface. The photothermal conversion efficiency of the properties of MXene. Du et al. similarly found that MXene increased photothermal conversion efficiency with a 9.4 °C higher surface temperature than PEG without MXene after solar irradiation.[80]

While LDHs, core/shell structures, metal additives, and other dopants are not well-studied, such materials have a high potential to enhance the PCM performance of PEG. For host materials, the maximum PEG encapsulation without leakage is key to high thermal storage capacities. PEG encapsulation may be increased by changing functional groups, pore sizes, or matrix morphologies. For dopants used to increase thermal conductivity, a high natural thermal conductivity and compatibility with PEG are important. Dopants should not interfere significantly with the crystallinity of PEG and should be minimally used to maximize the latent heat from high PEG content. The performance of PEG-based composite PCMs is summarized in Table 6.

4. Applications of PEG PCMs

This section overviews the critical properties of PEG-based PCMs for the most common and promising PCM applications, including optimal temperature ranges, flammability, toxicity, and thermal stability. Figure 8 summarizes the phase change temperature ranges, features, and challenges of promising PEG-based PCM implementations: thermal regulation systems in buildings, pavements, batteries, and smart textiles and TES applications like solar TES and movable TES.

Table 6. Summary of the phase change properties of other PEG-based composite PCMs.

Composite	Composition	Synthesis	T_m (°C)	T_c (°C)	ΔH_m (J/g)	ΔH_c (J/g)	Thermal conductivity (W/m·K)	Ref.
PEG/KH-LDH	55 wt% PEG	Solution impregnation	50.1	28.8	100.9	100.0	–	[146]
PEG/FLN-TiO_2	50 wt% PEG	Physical blending & impregnation	53.6	20.1	93.7	91.1	–	[147]
PEG/Ni foam	80 wt% PEG	Impregnation	65.9	41.0	132.1	135.8	0.64	[149]
PEG/TiO_2 foam	70 % impregnation ratio	Vacuum impregnation	64.1	145.9	36.3	142.9	0.51	[150]
PEG/Fe-MOG-100	92 wt% PEG	In-situ assembly/gelation	52.6	19.5	152.9	134.7	–	[152]
PEG/MOG	98 wt% PEG	Hydrothermal	68.0	39.0	158.4	165.3	0.33	[153]
PEG/Ag-GNS	98 wt% PEG	Ultrasonication & vacuum drying	60.2	35.6	177.2	179.4	0.32	[154]
PEG-AgNW/EV	78 wt% PEG-Ag	Physical blending & impregnation	60.0	43.0	112.1	114.8	0.68	[156]
PEG/BN/GNP	94 wt% PEG	Blending & Vacuum drying	63.6	40.3	168.3	160.4	0.65	[164]
PEG/CNIC	60 wt% PEG	Blending & impregnation	43.8	12.0	45.8	42.7	–	[166]
PEG/carbon/MXene	82 wt% PEG	Vacuum impregnation	51.3	38.9	140.0	135.6	–	[167]

Acronyms: KH-LDH (3-aminopropyl triethoxysilane-modified layered double hydroxide); FLN-TiO_2 (flower-like nanostructured TiO_2); Fe-MOG-100 (iron-benzenetricarboxylate metal-organic gel); MOG (metal-organic gel); Ag-GNS (graphene nanosheets doped with Ag nanoparticles); AgNW (Ag nanowires); EV (expanded vermiculite); BN (boron nitride); GNP (graphene nanoplatelets); CNIC (carbon nitride intercalation compounds).

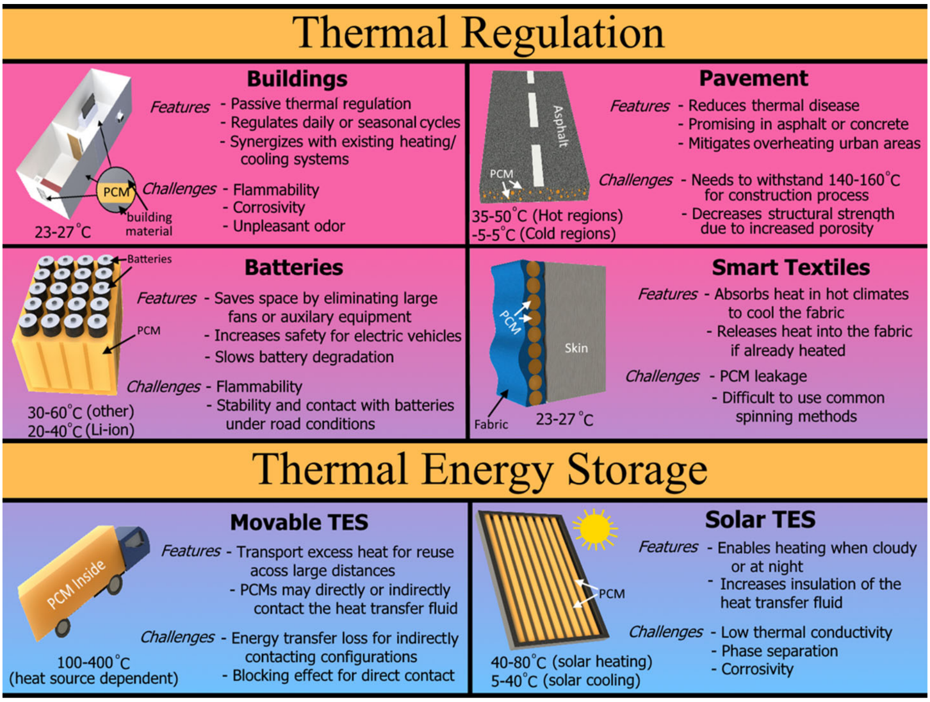

Figure 8. Summary of the optimal phase change temperature ranges, features, and challenges of promising PEG-based PCM applications.

4.1. Buildings

One of the most promising implementations of PCMs is the thermal management of buildings *via* active or passive methods.[168] PEG is an ideal choice as a PCM for building applications owing to its low toxicity and ideal phase change temperature within the range comfortable for people (20–30 °C).[169] As illustrated in Figure 9, PEG-based PCMs have been embedded directly into mortar,[170] concrete,[171] and wood[172,173] or into pipes embedded in the walls or floor.[169] Solar thermal energy can be stored up during the day and released when it is colder at night, reducing heating costs during colder seasons. In hotter seasons, PCM thermal management utilizes free cooling, which refers to cooling from a natural cold source, such as nighttime air, instead of artificially generated cold sources like air conditioning.[168] PCMs in the walls of a building reduce heat transfer between the outdoor environment and the indoor climate, resulting in more stable and comfortable ambient temperatures indoors.[174] PCMs can be used in conjunction with active heating/cooling systems, improving the thermal efficiency of the existing systems.[168] PCMs may also be indirectly used for the thermal management of buildings by increasing the efficiency of heat transfer fluids (HTFs) that recycle the heat in residential homes.[175,176]

However, PEG PCMs are slightly flammable,[174] so modifications that endow anti-flammable and anti-corrosive properties are required. For example, embedding the PEG PCM into an anti-flammable carbon skeleton[125] can significantly lower the risk of rapid combustion. Additionally, PEG-embedded concrete often has reduced physical strength,[169] which must be addressed for greater adoption in the building industry.

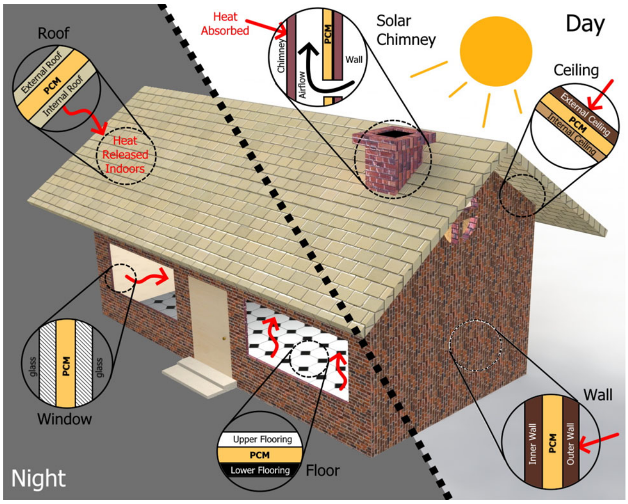

Figure 9. Schematic illustrating the application of PCMs in various components of a residential building for passive temperature regulation *via* heat absorption during the day and heat release during the night.

4.2. Pavements

Pavement materials, such as asphalt or concrete, crack under high heat due to oxidation and decreased stiffness, resulting in hazardous fissures for vehicles and people.[169,177] Freeze-thaw cycles and additional environmental stressors, such as snow melting salts and snowplowing, also weaken pavements.[177] The surface of pavements can heat up as high as 45 °C in the summer,[178] and dark asphalt may reach even higher temperatures of 60 °C, absorbing heat from solar irradiation, radiation from buildings, and friction from vehicles.[179] The higher surface temperatures contribute to higher air temperatures and the urban heat island effect, resulting in increased pollution, energy consumption, and clean water use.[180] PEG is an excellent candidate as pavement PCMs because of their suitable phase change temperature ranges between 30–60 °C,[181,182] tunable by PEG molecular weight. PEG-based PCMs can also improve resistance against moisture-induced damage.[183] Embedding PCMs into pavements reduces the temperature of the pavement because energy from sunlight is stored as latent heat in the PCMs, reducing pavement cracking from high temperatures and decreasing the urban heat island effect.[144] Moreover, the PEG PCMs release the stored heat to provide an anti-freezing effect, reducing destructive freeze-thaw cycles.

Three areas that need to be significantly improved are its compatibility with pavement mixtures, the thermal stability of PEG, and its thermal conductivity. PEG-embedded pavements generally have weaker physical strength than PCM-less pavements. PEG-based composite PCMs must be highly form-stable during phase change to limit the weakening effect.[179,184] Moreover, encapsulating PEG with strong and thermally diffusive materials like expanded graphite[126,185] and controlling the PCM content in pavement blends reduce the weakening effects of embedded PEG PCMs.[186] Pavements with smaller PEG PCM sizes relative to asphalt have reported greater mechanical strength.[187] Second, PEG-based PCMs must withstand high temperatures around 140–160 °C, which are required during the mixing and construction process of pavement materials.[177] Lastly, PEG-based PCMs must have high thermal conductivity to distribute heat from the surface quickly to PCMs deeper in the pavement. Issues regarding the thermal stability of PEG PCMs and their thermal conductivity can be addressed by adding thermally resistant host materials[149,188] and thermally conductive fillers.[189]

4.3. Batteries

PCMs in battery applications regulate the thermal stress on batteries, slowing the degradation of the battery.[190] Traditionally, battery cooling has been performed with fans or other auxiliary hardware, which takes up significant space and adds to the cost of battery implementation. Recently, PCMs have been investigated to absorb excess battery heat to prevent thermal runaway.[191] As batteries are being increasingly used in electric vehicles (EVs) and various IoT devices, PCMs have become promising options for thermal regulation.[192] PEG-based PCMs are excellent candidates for battery application, with an ideal phase transition temperature between 20–40 °C.[193] Moreover, the low flammability of PEG makes it a promising choice to reduce or prevent fires caused by thermal runaways. Multifunctional PEG-based PCMs may also endow electromagnetic interference shielding, electrical conductivity, and other functions, increasing the battery's compatibility with other electrical components.[194]

However, the thermal conductivity of PEG must be significantly increased with conductive fillers or encapsulating materials to increase the heat dissipation rate, especially in passively cooled battery systems.[117,195,196] For EV implementations, form-stable PEG-based composite PCMs are required because using external containers that add to the vehicle's weight and decrease the vehicle's total usable volume.[197,198]

4.4. Smart textiles

A more recent application of PCMs is in smart textiles for improved human thermoregulation.[199] Low MW PEG-based PCMs have ideal phase transition temperatures near the human body temperature (23–27 °C) and are nontoxic, making them excellent candidates for thermoregulating fabrics. Moreover, PEG-based PCMs can be easily incorporated into smart textiles *via* coating,[200] lamination,[201] or spinning methods.[199] Figure 10(a) illustrates how PEG-based composite PCMs can help reduce skin temperatures in hot climates by absorbing heat from the body and environment. When moving to a colder environment, the heat from the PCM may be released to warm the body (Figure 10b). For example, the insulating ability of cotton fabric can be increased. A PEG/lignin-coated cotton fabric took 4 times longer to reach 60 °C than the control cotton fabric because of the latent heat absorption process.[200] Delaying the onset of heat or cold increases the insulation of textiles, which is ideal for resisting sudden temperature changes.

The largest issues with PEG-based PCMs for smart textile applications are the integration with fabric material and stability after repeated wash cycles.[114,202] Spinning methods have been more popular than coating and lamination methods for fabricating thermoregulating smart textiles because the spinning route stably embeds the PCM

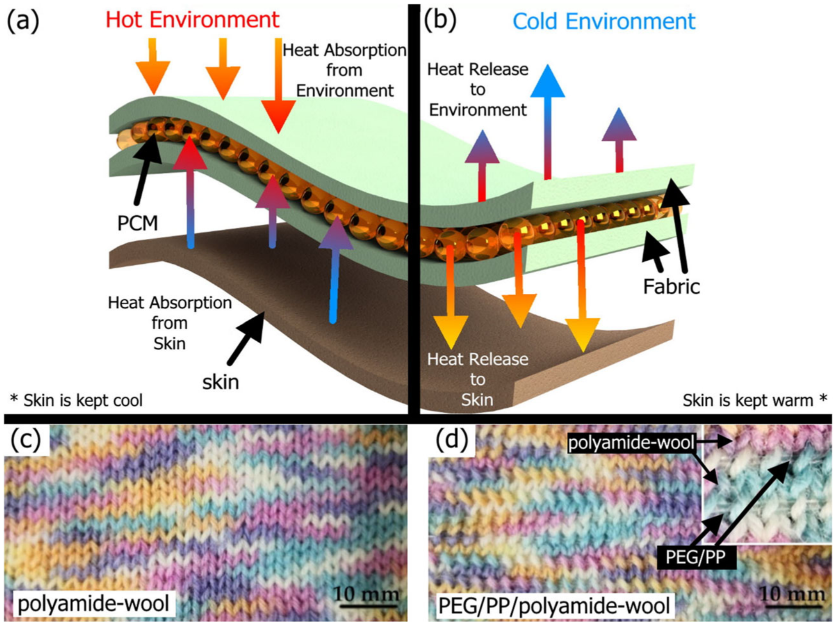

Figure 10. Heat absorption and release from PCM-embedded fabrics on skin in (a) hot and (b) cold environments, generating a stable microclimate between the fabric and skin. (c) Knitted polyamide-wool fabric and (d) PEG/polypropylene (PEG/PP)/polyamide-wool fabric.[203] (Adapted with permission from [203], Copyright MDPI, 2021).

capsules within the fibers, making them less visible and more resistant to high-stress machine washing.[203] Comparing fabric without PCM (Figure 10c) to fabric with PCM (Figure 10d) shows how well-hidden PEG PCMs can be when integrated using spinning methods. Form-stable PEG-based composites must be used to resist leaking during hot-water washing.[204] Because PEG is a water-soluble polymer,[205] PEG must also be strongly encapsulated to prevent leaching into the laundry. Moreover, spinning methods have been developed to yield PEG-based PCM fibers without harmful and expensive organic solvents.[206] PEG PCMs doped with carbon materials or MXenes are also promising for textile applications that require high photothermal conversion efficiencies.[207]

4.5. Solar TES

The benefit of solar-heated PEG PCMs has already been discussed for thermal regulation applications in buildings (Sec. 4.1), pavements (Sec. 4.2), and even textiles (Sec. 4.4). In this section, solar TES refers to systems designed to maximize solar thermal energy harvesting, similar to how solar panels are designed to maximize photoelectric energy harvesting. Solar TES implementations are often combined with HTFs to store and distribute heat. PEG is a promising PCM for solar TES, owing to its appropriate transition temperature range between 40–80 °C and large latent heat.

For solar water heating, PEG PCMs can be placed under direct sunlight, similar to solar panels, but instead of converting solar irradiation into electrical energy, PCM-integrated solar thermal collectors convert the radiation into thermal energy (latent heat).

PEG PCMs can also be integrated into the HTF that passes through the solar thermal collectors. In either implementation, the PEG PCMs allow heat to be continuously transferred passively into the solar heater system, even with cloud coverage.[208]

However, the low thermal conductivity and photothermal conversion efficiency of PEG severely limit its viability in solar heating. Both issues have been simultaneously addressed in recent investigations by encapsulating PEG in thermally conductive host materials with high photothermal conversion efficiencies, such as various carbon materials (Sec. 3.2),[209] MXenes (Sec. 3.4),[210] and other thermally conductive additives.[211–213] Because PEG is a water-soluble polymer,[205] it can be easily integrated into the HTF to increase the energy density of the HTF. However, this requires the PEG-based PCMs to be encapsulated with strong guest-host interactions to prevent leakage in aqueous environments.[82] Recent work has also been done to use PEG-modified with thermally conductive nanofillers as an HTF.[159]

An exciting method of achieving solar water heating while reducing dependence on direct solar heating is using solar panels as the solar thermal collector. Photovoltaic (PV) cells that can reach around 60–80 °C in the summer need to be cooled since the most common silicon-based PV cells lose around 0.5% efficiency for every 1 °C increase above 25 °C.[214] Because low MW PEG-based PCMs have a constant temperature of around 30–40 °C during phase change, the PEG-based PCM can effectively maintain the temperature of PV cells. Sheik et al. achieved over 8% higher output and efficiency from standard silicon-based PV cells by implementing PEG-based composite PCMs.[214] In another experiment, low MW PEG-600 with a phase change 23–26 °C had a maximum of 18 °C lower temperature than a conventional PV cell.[215] However, the PEG PCM completely melted after 80 min and could not contribute to further cooling. Despite having a phase transition temperature within the optimal range for PV efficiency, lower MW PEG may not be as long-lasting as higher MW PEG, which has larger latent heat capacities and a slower melting rate.[216]

4.6. Movable TES

Movable-thermal energy storage (M-TES), also known as off-site waste heat recovery, allows industrial waste heat to be recycled at distant locations. By repurposing waste heat, M-TES can lower CO_2 emissions and enhance energy efficiency.[217] Form-stable PCMs are especially promising for M-TES applications because they do not need external containers, have minimal volumetric change during energy charge/discharge, and have extremely high energy densities. Waste heat is usually transferred to the PCMs *via* HTFs, and the PCMs may either be in direct contact or indirect contact with the HTFs.[218]

High MW PEG-based PCMs are excellent candidates for M-TES applications owing to their very high specific latent heat and low thermal conductivity, allowing large amounts of energy to be transferred for long durations without significant heat loss. Although low-grade industrial waste heat often has temperatures between 100–400 °C, above the temperature range of PEG, the temperatures may be decreased by the HTF to prevent thermal degradation.[219] PEG-based composite PCMs may be easily optimized to maximize their thermal conductivity for faster heat transfer between the HTF and

PCM instead of optimizing for long-distance travel. Higher thermal conductivity is more important for PEG-based PCMs heated indirectly by HTFs, as heat has to flow from the container holding the HTF to a separate container containing the PCMs.[218] Indirect PCMs, on the other hand, must have excellent form-stability to allow the solid-state PCMs to be easily separated from the HTFs.[220] Because HTFs are usually water or aqueous solutions, PEG should be encapsulated with hydrophobic shells to minimize leakage or modified with hydrophobic fillers to increase stability under aqueous conditions.

5. Future prospects and conclusion

PEG is a promising PCM due to its high latent heat and easily tunable phase change temperatures. To improve its low thermal conductivity and endow form-stability, PEG has been modified *via* doping and encapsulation with thermally conductive or porous materials, including polymers, porous carbon, graphene, CNTs, silica, LDHs, MXenes, MOFs, and metal foams. While PEG-based composite PCMs have been designed with high latent heat, thermal conductivity, photoconversion efficiency, and flame resistance, the thermal storage capacity, chemical stability, and mechanical strength must be improved.

- For thermally regulated buildings, it is necessary to develop novel methods of incorporating PEG-based PCMs directly into building materials like concrete and wood without significantly reducing structural integrity. Directly embedding PEG PCMs into construction materials will be cheaper than integrating a completely separate PCM system into construction using pipes or other indirect methods. Developing PEG-based PCMs with fire-resistant characteristics can also help reduce damage from fires.
- The mechanical strength of PEG-based PCMs embedded in pavements should be improved by optimizing PCM/pavement ratios and encapsulating PEG with high temperature-resistant materials. The thermal conductivity must be improved, and PEG leakage must be addressed to reduce thermal disease and increase the lifetime of pavements.
- PEG-based PCMs used to thermally regulate batteries must have higher thermal conductivities to spread and dissipate waste heat quickly without using fans. Moreover, PEG may be modified with anti-flammable characteristics to minimize the danger of thermal runaways. Multifunctional PEGs should also be investigated, as PEG can host various materials that endow electromagnetic shielding, electrical conductivity, etc.
- For thermally regulating textiles, there is a need for more facile, safe, environmentally friendly, and scaleable fabrication methods for embedding PEG into fibers. PEG-based PCMs must endure multiple wash cycles in high heat, requiring excellent form stability and mechanical strength, especially in aqueous environments.
- PEG-based composite PCMs are still limited in solar TES applications by their low thermal conductivity and photothermal energy conversion efficiencies. More

work should be done to increase the solar absorption of PEG and integrate thermally conductive fillers without significantly decreasing the high specific latent heat of PEG.
- M-TES systems with direct heat transfer between the HTF and PEG-based PCMs require the PCMs to be highly stable in the HTFs and easily separable from the HTFs. Hence, more development is needed in synthesizing form-stable PEG-based PCMs under aqueous conditions.
- Emerging PCM-integrated technologies such as smart drug delivery, nanosensors, and other electronics are also promising areas for further investigation.

In conclusion, PEG-based composite PCMs have been developed to have form stability, reduced leakage, enhanced thermal conductivity, and higher photothermal energy conversion efficiency, amongst other improvements. Encapsulating PEG with various polymers, especially cross-linked polymers, yields excellent form stability with minimal leakage due to strong intermolecular interactions and optimized pore sizes. The fabrication of PEG/polymer nanofibers *via* spinning methods was a common strategy for smart textiles. Carbon composites are promising due to their sustainable origin, abundance, low cost, and nontoxic byproducts. CNTs are highly conductive dopants that increase the photothermal conversion efficiency of PEG. Their facile functionalizability allows PEG/CNT composite PCMs to have finely tuned thermal properties. Graphene and its various derivatives have been primarily used as additives that can increase the thermal conductivity of PEG many times with less than 2 wt% loading. PEG can also be immobilized between the layers of graphene to reduce leakage. Biologically derived, hierarchically porous carbons are excellent candidates for entrapping PEG, owing to their interconnected pore structure, high surface area, and high thermal conductivity. PEG/silica composites have some of the highest PEG encapsulation efficiencies because of their highly porous structures. PEG/silica PCM's excellent thermal stability makes them suitable for thermally demanding applications like in pavements and movable TES. Other promising host and filler materials, including LDHs, MOFs, metal foams, metallic dopants, and MXenes, have shown good form stability and enhanced thermal conductivity but require further investigation.

Disclosure statement

The authors declare that there are no relevant financial or non-financial competing interests.

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